



國立成功大學
National Cheng Kung University

Introduction to Emerging Memory Techniques

Lecture 2: Phase Change Memory

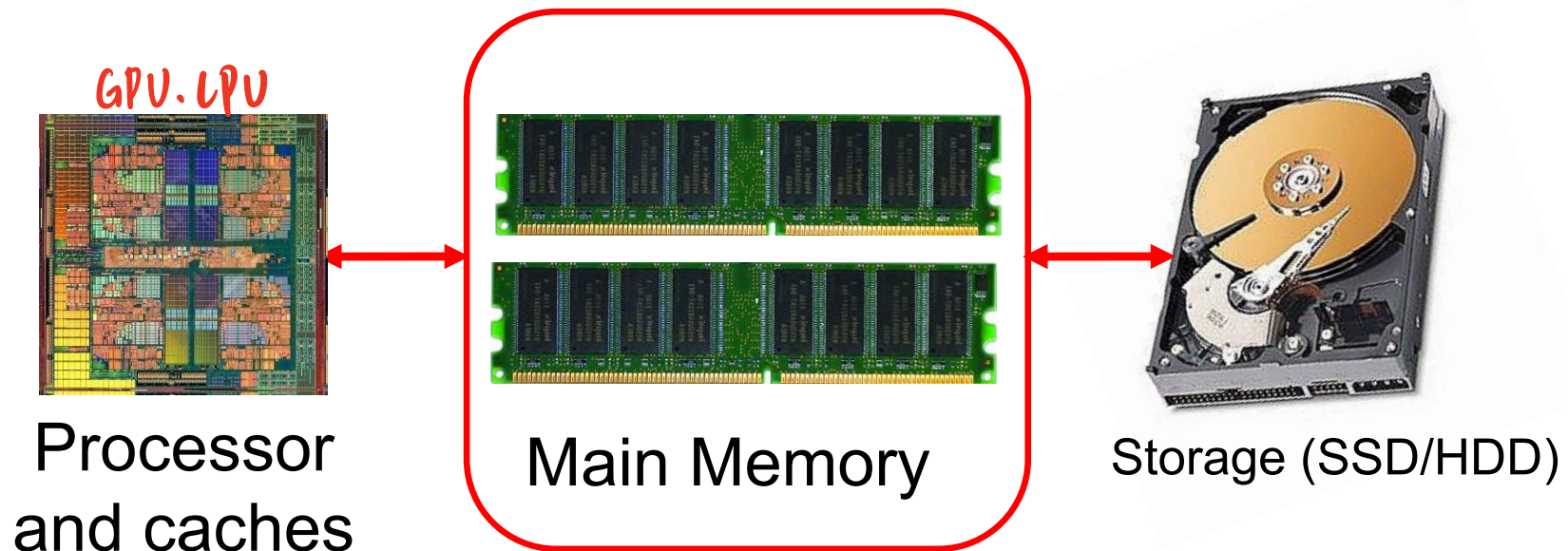
Prof. Chien-Chung Ho



Agenda

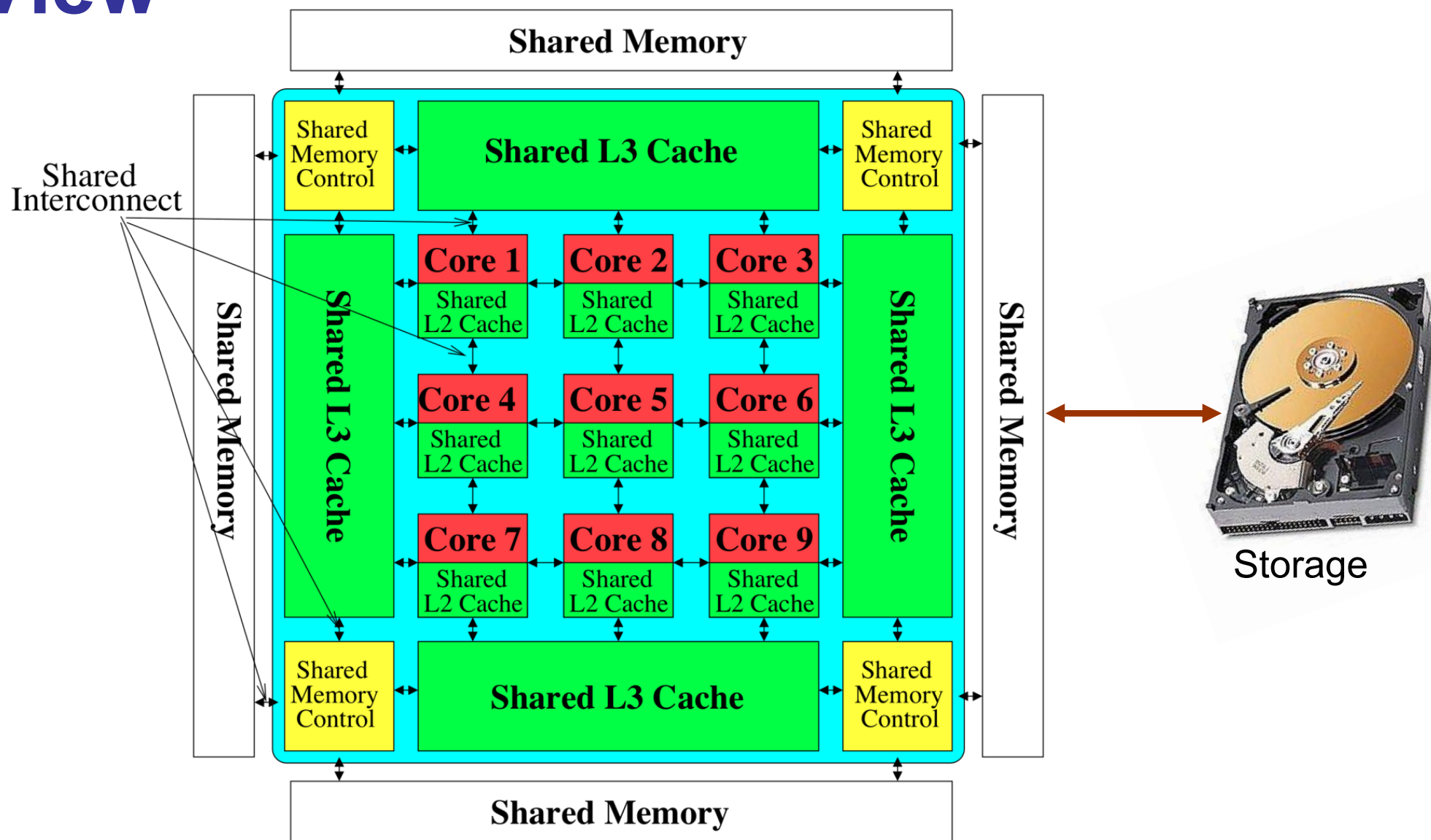
- Overview on the main memory system
- Opportunity: Emerging Memory Technologies
 - Background
 - Phase Change Memory
 - PCM (or Technology X) as DRAM Replacement
 - Hybrid Memory Systems
- Conclusion

The Main Memory System



- **Main memory is a critical component of all computing systems:** server, mobile, embedded, desktop, sensor
- **Main memory system must scale** (in size, technology, efficiency, cost, and management algorithms) to maintain performance growth and technology scaling benefits

Memory System: A *Shared Resource* View



State of the Main Memory System

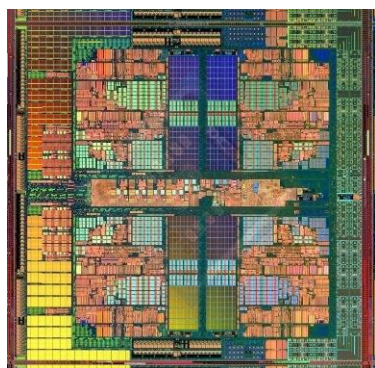
- Recent technology, architecture, and application trends
 - lead to new requirements
 - E.g., data sharing over L1 caches
 - exacerbate old requirements
 - Caching policy, scalability, performance, etc.
- **DRAM and memory controllers**, as we know them today, are (will be) **unlikely to satisfy all requirements**
- Some **emerging non-volatile memory technologies** (e.g., PCM) **enable new opportunities**: memory+storage merging
- We need to rethink the main memory system
 - to fix DRAM issues and enable emerging technologies
 - to satisfy all requirements

Major Trends Affecting Main Memory

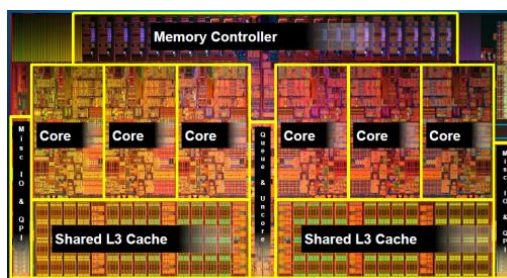
- Need for main memory capacity, bandwidth, QoS increasing
 - **Multi-core**: increasing number of cores
 - **Data-intensive applications**: increasing demand/hunger for data
 - **Consolidation**: cloud computing, GPUs, mobile
- Main memory energy/power is a key system design concern
- DRAM technology scaling is ending

Example Trend: Many Cores on Chip

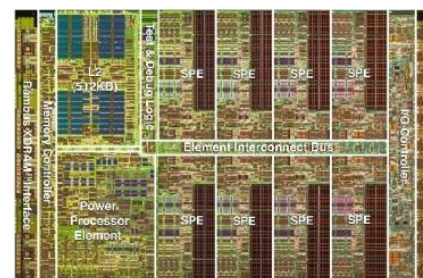
- Simpler and lower power than a single large core
- Large scale parallelism on chip



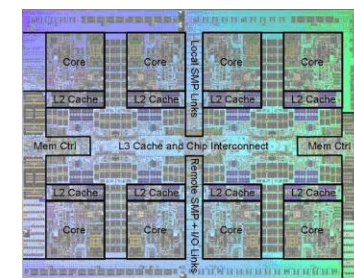
AMD Barcelona
4 cores



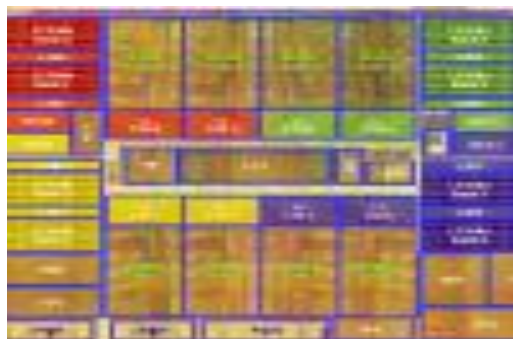
Intel Core i7
8 cores



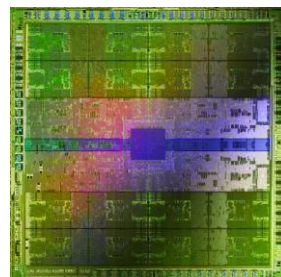
IBM Cell BE
8+1 cores



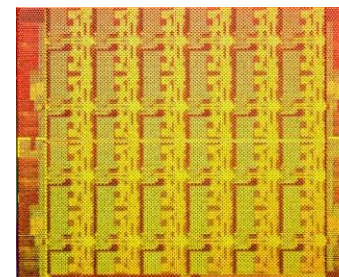
IBM POWER7
8 cores



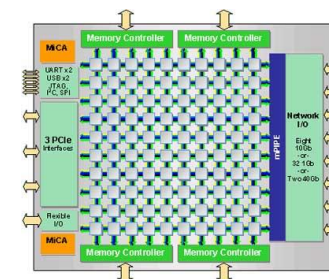
Sun Niagara II
8 cores



Nvidia Fermi
448 "cores"



Intel SCC
48 cores, networked

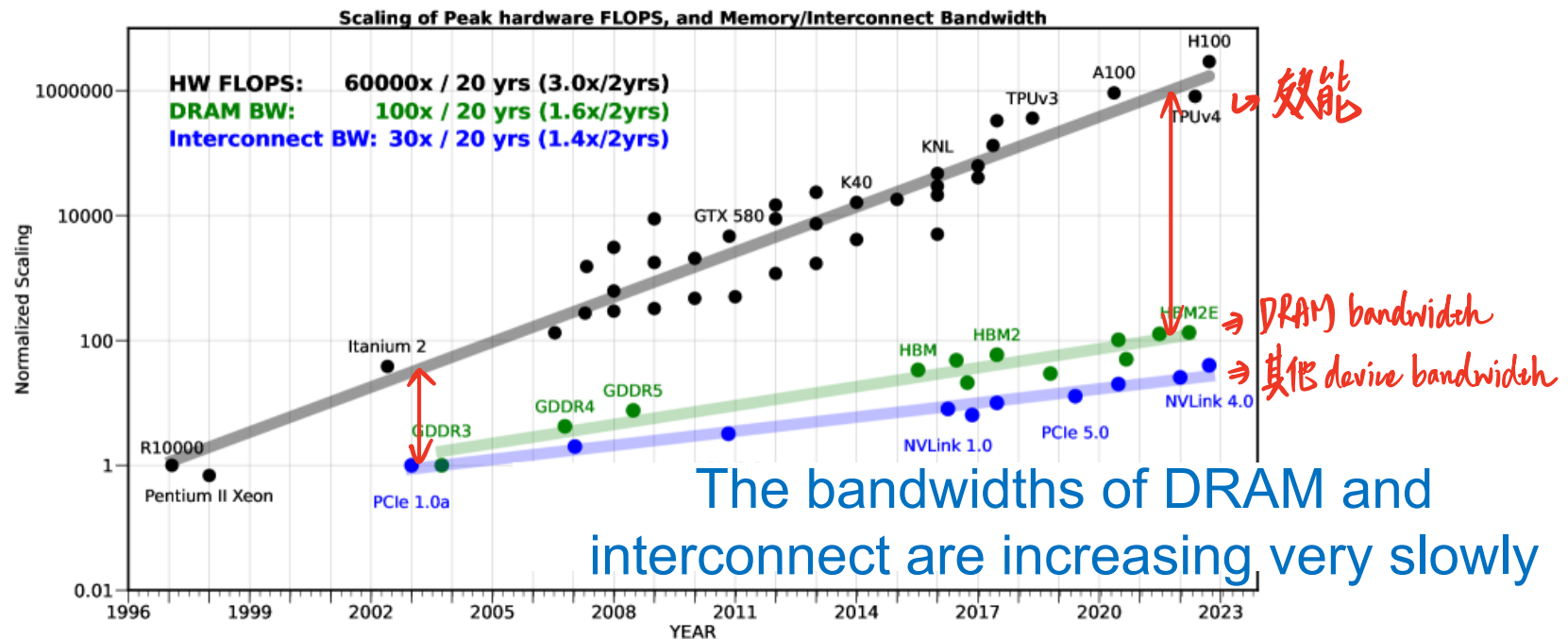


Tiler TILE Gx
100 cores, networked

Consequence: The Memory Capacity Gap

Core count doubling ~ every 2 years

DRAM DIMM capacity doubling ~ every 3 years



- Memory capacity per core expected to drop by 30% every two years
- Trends worse for memory bandwidth per core!
- Memory wall problem
 - The time to complete an operation is dependent on how fast we can perform the arithmetic as well as how fast we can feed data to the arithmetic units of hardware.
 - No matter how fast the hardware could perform arithmetic ops per second, the problem will be entirely limited by DRAM bandwidth.

Major Trends Affecting Main Memory

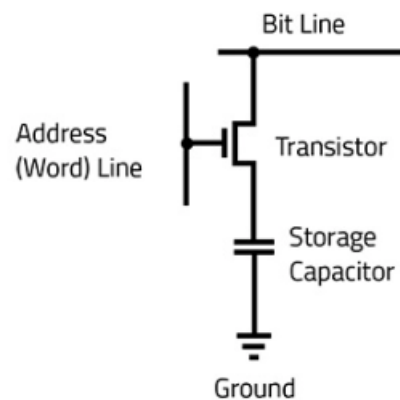
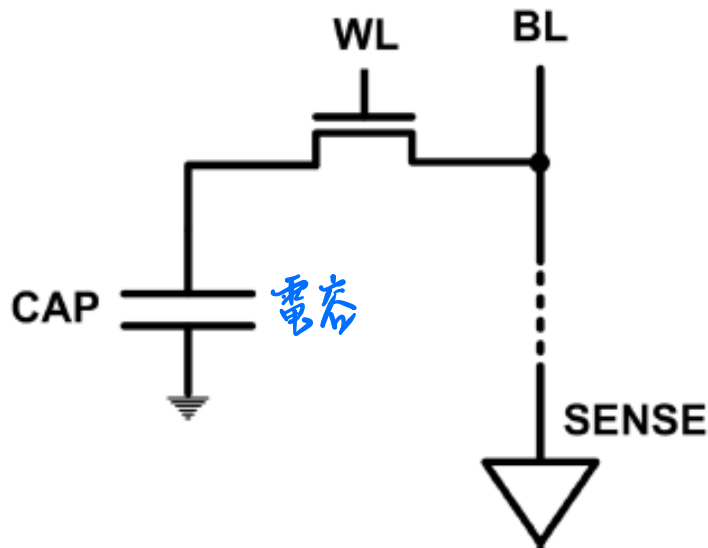
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 - **Multi-core**: increasing number of cores
 - **Data-intensive applications**: increasing demand/hunger for data
 - **Consolidation**: cloud computing, GPUs, mobile
- Main memory energy/power is a key system design concern
 - ~40-50% energy spent in off-chip memory hierarchy [Lefurgy, IEEE Computer 2003] ↳ 耗能在搬data上
 - DRAM consumes power even when not used (periodic refresh)
- DRAM technology scaling is ending

Major Trends Affecting Main Memory

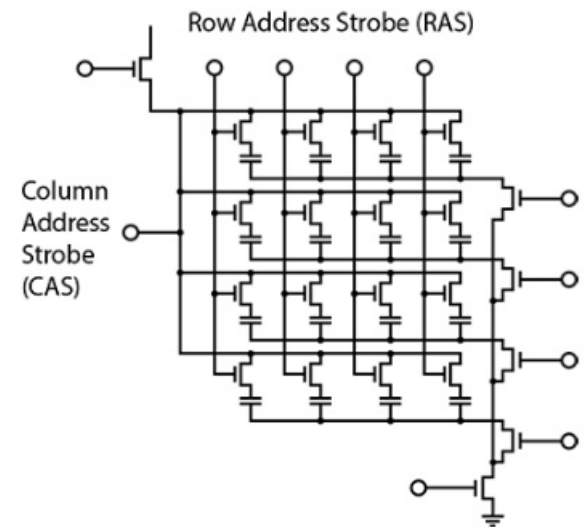
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 - **~40-50% energy spent in off-chip memory hierarchy** [Lefurgy, IEEE Computer 2003]
 - **DRAM consumes power even when not used (periodic refresh)**
- DRAM technology scaling is ending
 - ITRS projects **DRAM will not scale easily below X nm**
 - Scaling has provided many benefits:
 - **higher capacity (density), lower cost, lower energy**

The DRAM Scaling Problem

- DRAM stores charge in a capacitor (charge-based memory)
 - Capacitor must be large enough for reliable sensing
 - Access transistor should be large enough for low leakage and high retention time
 - Scaling beyond 40-35nm (2013) is challenging [ITRS, 2009]



Single Memory Cell

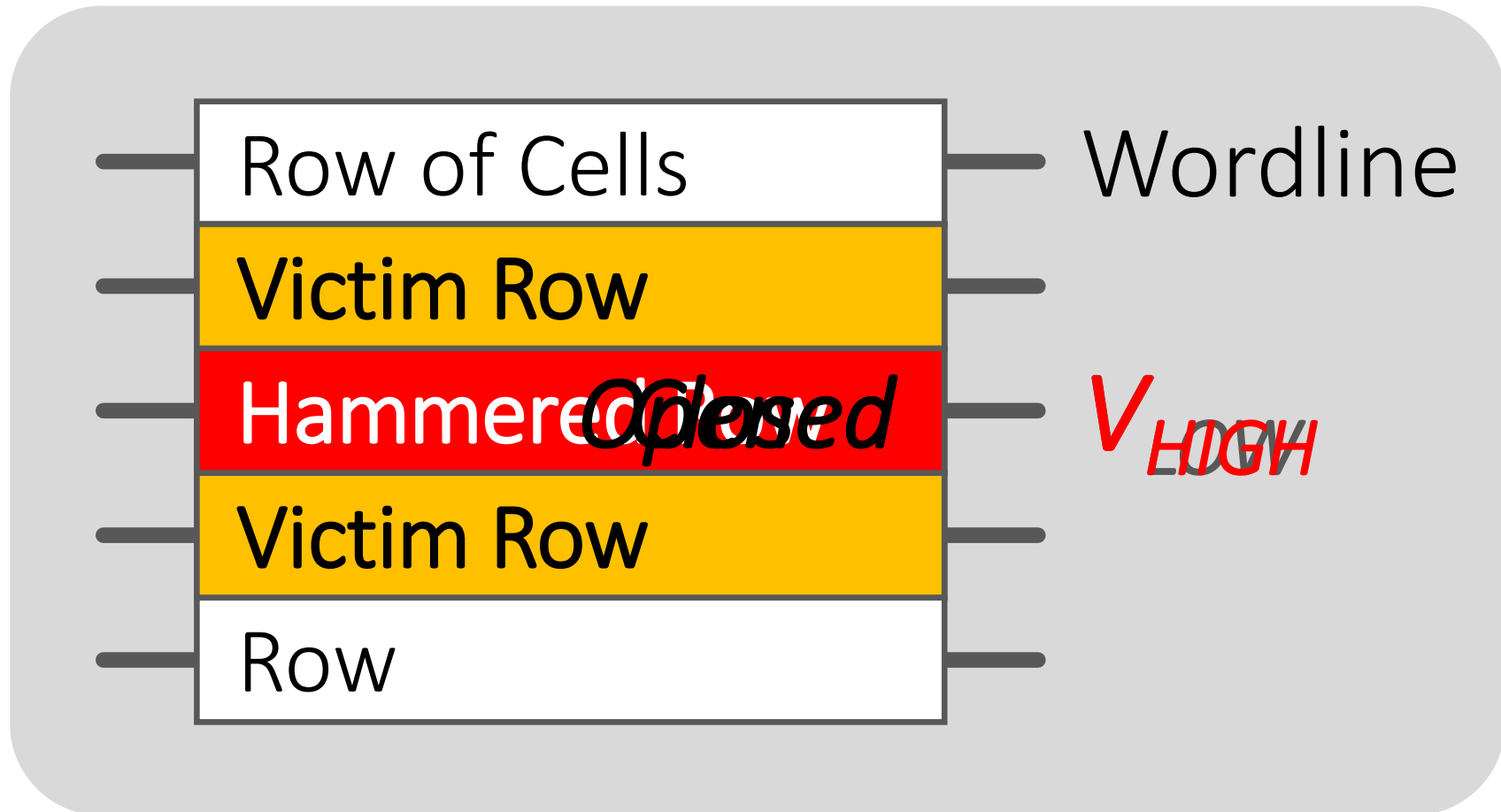


Memory Cell Array

How DRAM cells perform read and write: <https://www.electfans.com/v/30922/>

- DRAM's capacity, cost, and energy/power are hard to scale

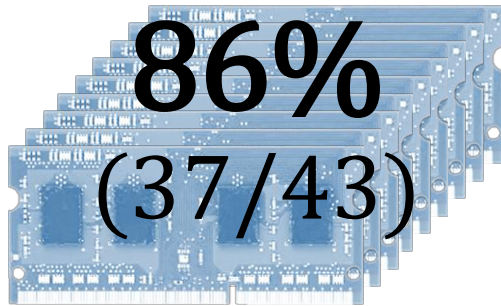
An Example of The Scaling Problem



Repeatedly opening and closing a row induces disturbance errors in adjacent rows in most real DRAM chips [Kim+ ISCA 2014]

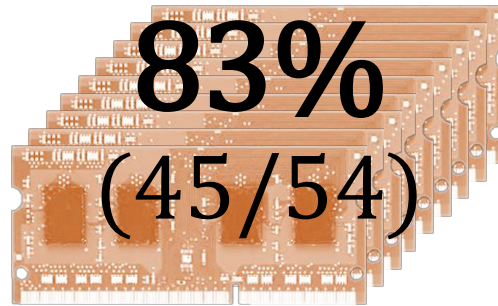
Most DRAM Modules Are At Risk

A company



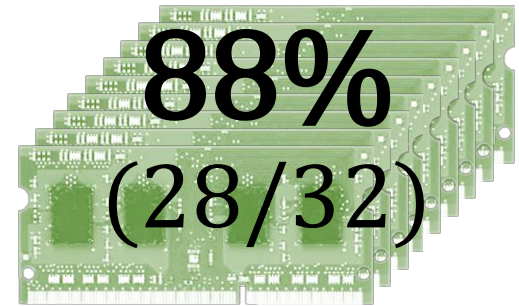
Up to
 1.0×10^7
errors

B company



Up to
 2.7×10^6
errors

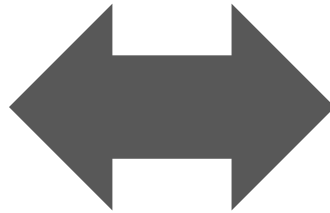
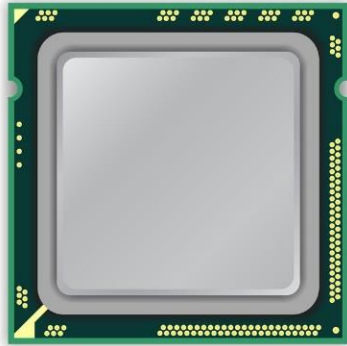
C company



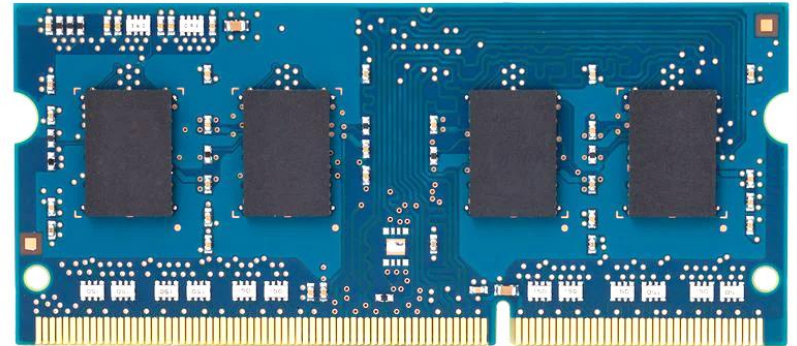
Up to
 3.3×10^5
errors

Kim+, "Flipping Bits in Memory Without Accessing Them: An Experimental Study of DRAM Disturbance Errors," ISCA 2014.

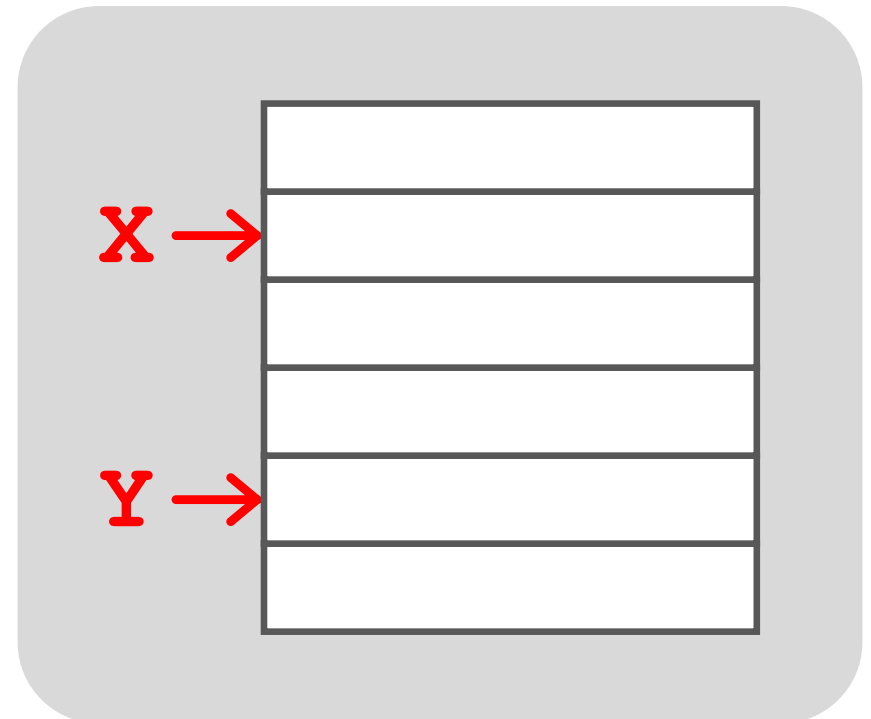
x86 CPU



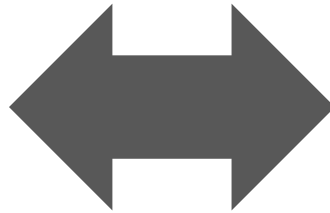
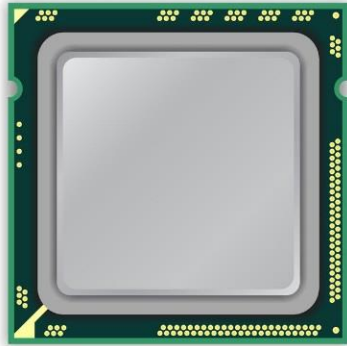
DRAM Module



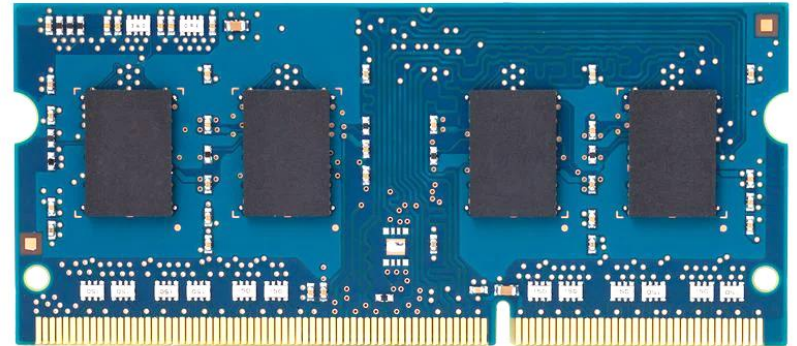
```
loop:  
  mov  (X), %eax  
  mov  (Y), %ebx  
  clflush (X)  
  clflush (Y)  
  mfence  
  jmp  loop
```



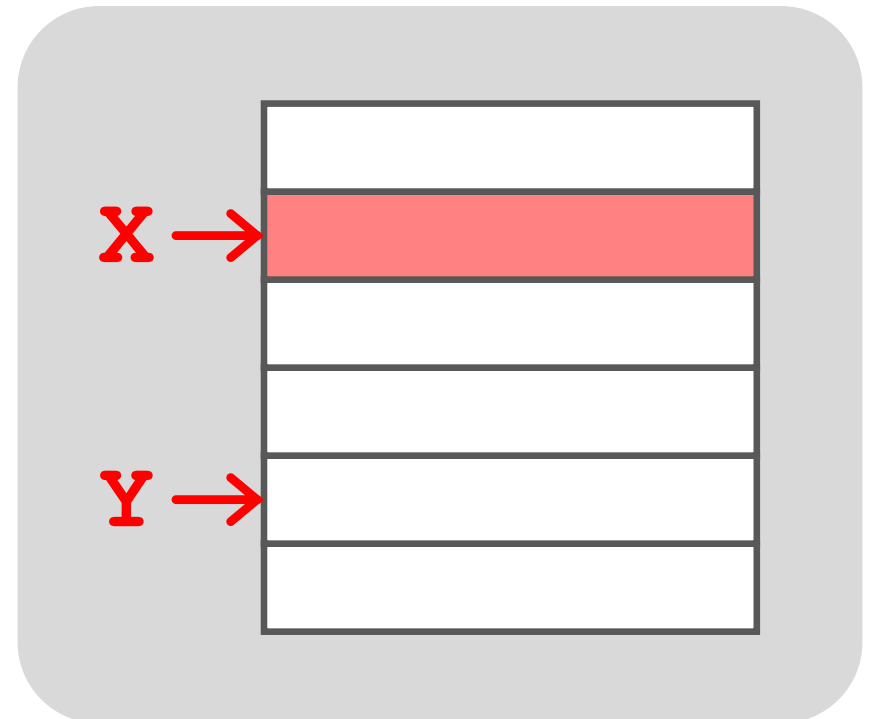
x86 CPU



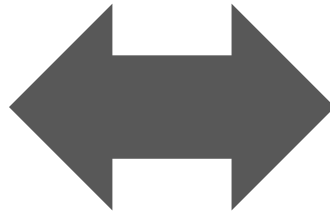
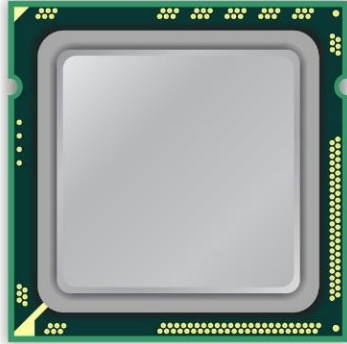
DRAM Module



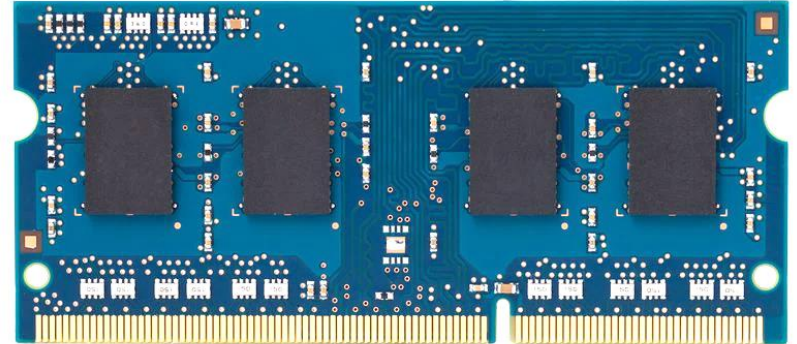
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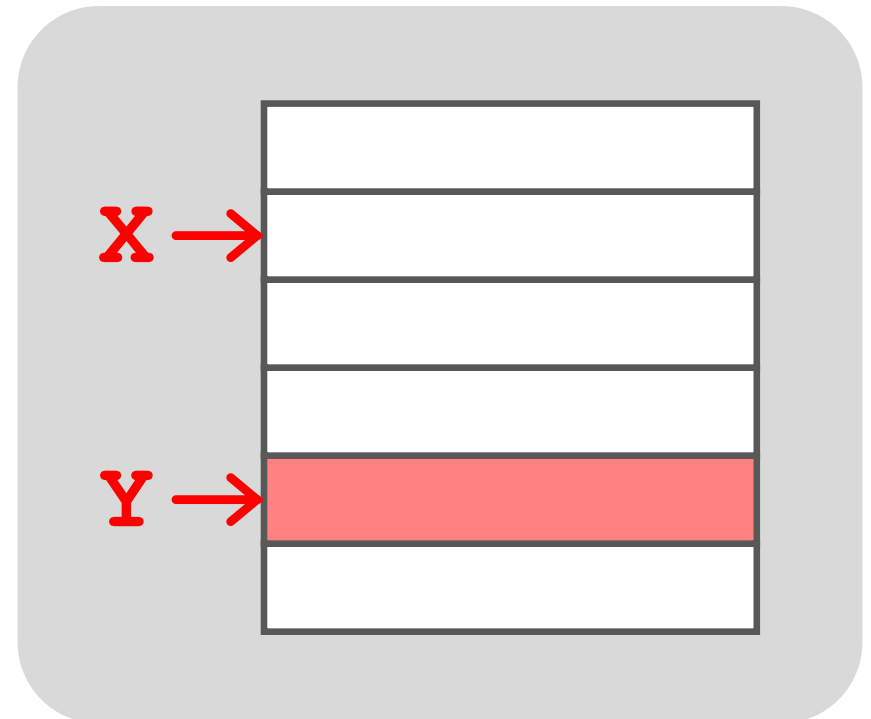
x86 CPU



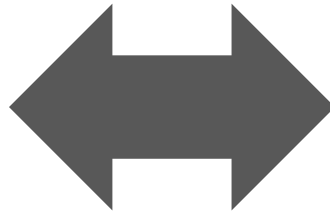
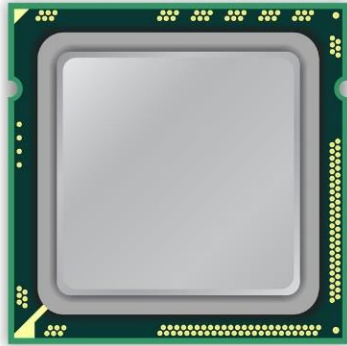
DRAM Module



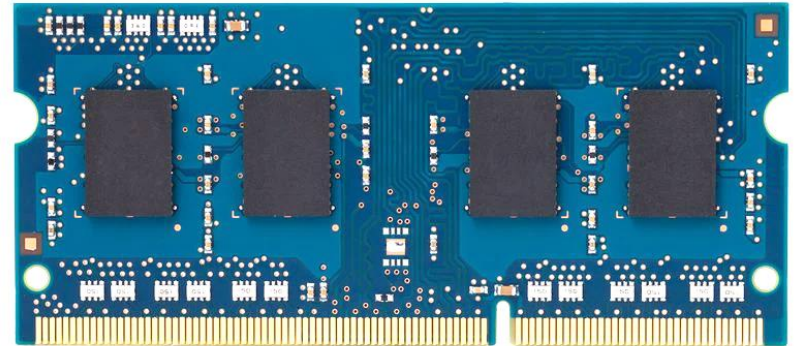
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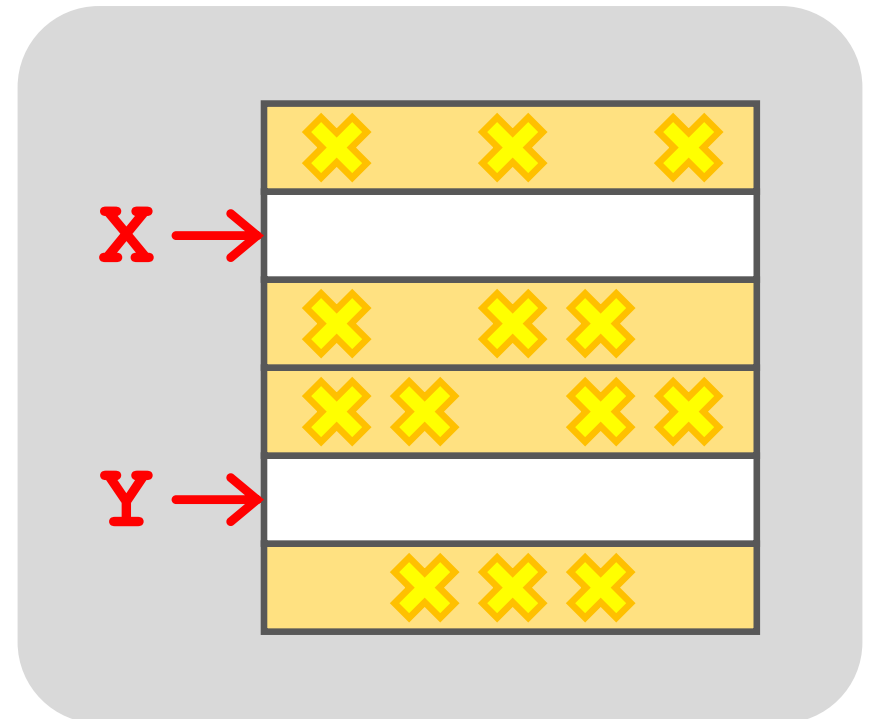
x86 CPU



DRAM Module



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```



Observed Errors in Real Systems

CPU Architecture	Errors	Access-Rate
Intel Haswell (2013)	22.9K	12.3M/sec
Intel Ivy Bridge (2012)	20.7K	11.7M/sec
Intel Sandy Bridge (2011)	16.1K	11.6M/sec
AMD Piledriver (2012)	59	6.1M/sec

- *A real reliability & security issue*
- *In a more controlled environment, we can induce as many as **ten million** disturbance errors*

Requirements from an Ideal Memory System

- Traditional
 - Enough capacity
 - Low cost
 - High system performance (high bandwidth, low latency)
- New
 - Technology scalability: lower cost, higher capacity, lower energy
 - Energy (and power) efficiency
 - QoS support and configurability (for consolidation)

Requirements from an Ideal Memory System

- Traditional

- Higher capacity
- Continuous low cost
- High system performance (higher bandwidth, low latency)

- New

- Technology scalability: lower cost, higher capacity, lower energy
- Energy (and power) efficiency
- QoS support and configurability (for consolidation)

Emerging, resistive memory technologies (NVM) can help

Solutions to the DRAM Scaling Problem

- Two potential solutions
 - Tolerate DRAM (by taking a fresh look at it)
 - Error Detection and Correction
 - Fault-Tolerant Design
 - Hardware Monitoring
 - Degradate Mode Operation
 - Fault Isolation
 - Enable emerging memory technologies to eliminate or minimize DRAM
- Do both
 - Hybrid memory systems
 - E.g., DRAM + PCM hybrid memory systems

Solution 1: Tolerate DRAM

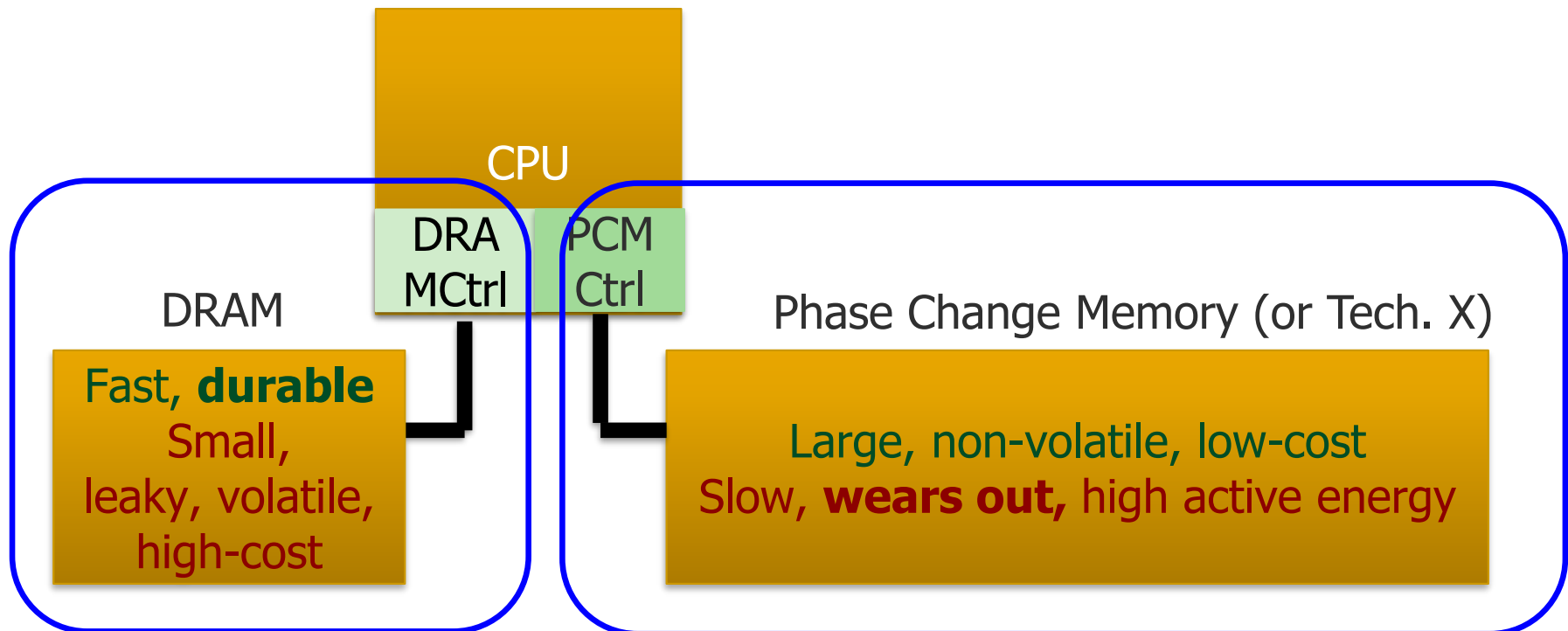
- Overcome DRAM shortcomings with
 - System-DRAM co-design
 - Novel DRAM architectures, interface, functions
 - Better waste management (efficient utilization)
 - Key issues that we need to tackle
 - Reduce refresh energy
 - Improve bandwidth and latency
 - Reduce waste
 - Enable reliability at low cost
-
- Liu, Jaiyen, Veras, Mutlu, “RAIDR: Retention-Aware Intelligent DRAM Refresh,” ISCA 2012.
 - Kim, Seshadri, Lee+, “A Case for Exploiting Subarray-Level Parallelism in DRAM,” ISCA 2012.
 - Lee+, “Tiered-Latency DRAM: A Low Latency and Low Cost DRAM Architecture,” HPCA 2013.
 - Liu+, “An Experimental Study of Data Retention Behavior in Modern DRAM Devices” ISCA’13.
 - Seshadri+, “RowClone: Fast and Efficient In-DRAM Copy and Initialization of Bulk Data,” 2013.

Solution 2: Emerging Memory Technologies

- Some emerging resistive memory technologies seem more scalable than DRAM (and they are non-volatile)
- Example: Phase Change Memory
 - Expected to scale to 9nm (2022 [ITRS])
 - Expected to be denser than DRAM: can store multiple bits/cell
- But, emerging technologies have shortcomings as well
 - Can they be enabled to replace/augment/surpass DRAM?

- Lee, Ipek, Mutlu, Burger, “[Architecting Phase Change Memory as a Scalable DRAM Alternative](#),” ISCA 2009, CACM 2010, Top Picks 2010.
- Meza, Chang, Yoon, Mutlu, Ranganathan, “[Enabling Efficient and Scalable Hybrid Memories](#),” IEEE Comp. Arch. Letters 2012.
- Yoon, Meza et al., “[Row Buffer Locality Aware Caching Policies for Hybrid Memories](#),” ICCD 2012 Best Paper Award.

Hybrid Memory Systems



Hardware/software manage data allocation and movement
to achieve the best of multiple technologies

Meza+, "Enabling Efficient and Scalable Hybrid Memories," IEEE Comp. Arch. Letters, 2012.
Yoon, Meza et al., "Row Buffer Locality Aware Caching Policies for Hybrid Memories," ICCD 2012 Best Paper Award.

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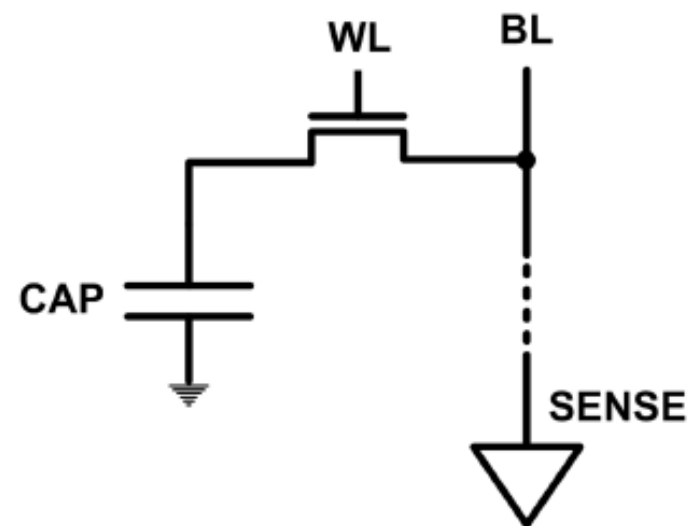
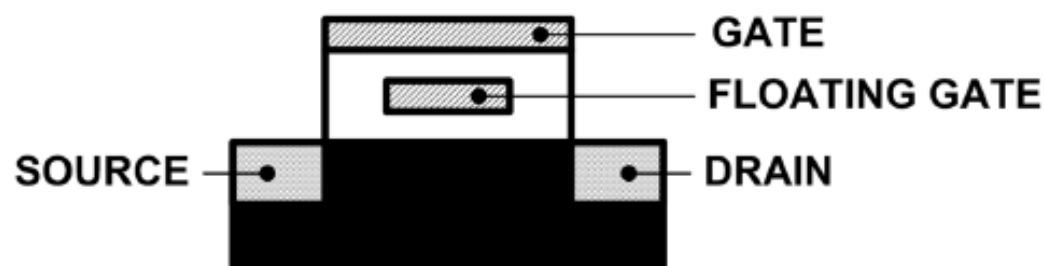
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Charge vs. Resistive Memories

- Charge Memory (e.g., DRAM, Flash)
 - Write data by capturing charge Q
 - Read data by detecting voltage V
- Resistive Memory (e.g., PCM, STT-MRAM, memristors)
 - Write data by pulsing current dQ/dt
 - Read data by detecting resistance R

Limits of Charge Memory

- Difficult charge placement and control
 - Flash: floating gate charge
 - DRAM: capacitor charge, transistor leakage
- Reliable sensing becomes difficult as charge storage unit size reduces



Emerging Resistive Memory Technologies

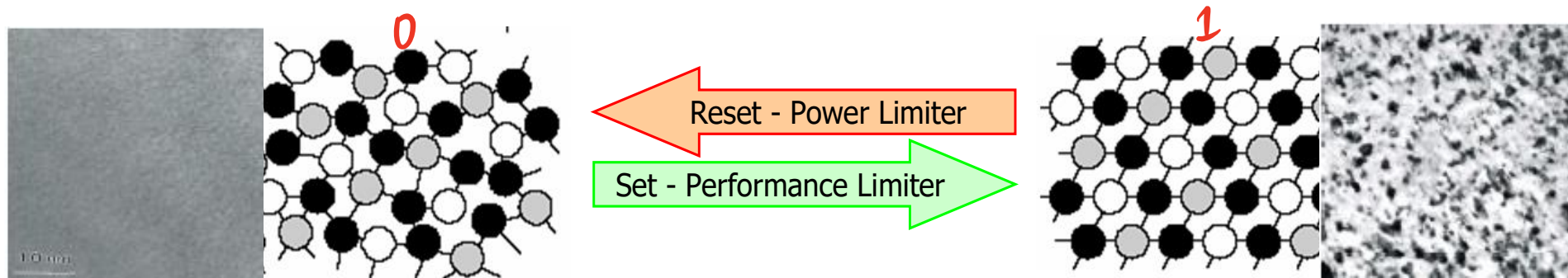
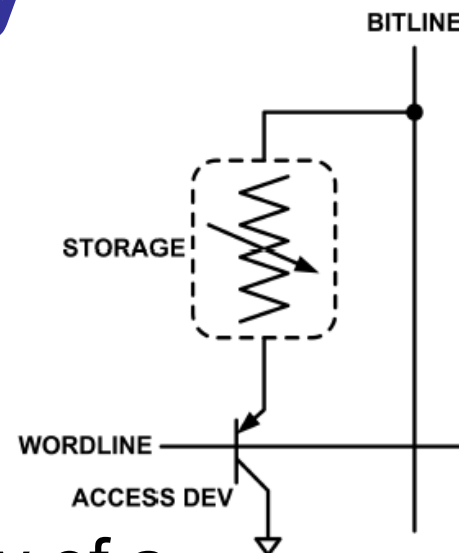
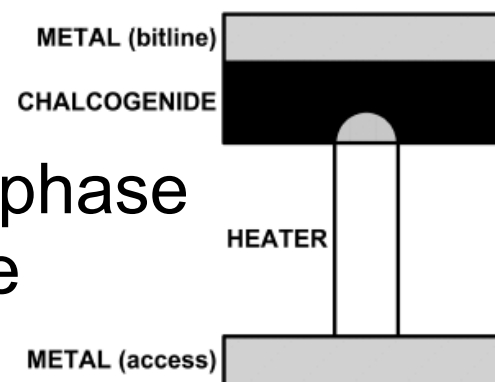
- PCM
 - Inject current to change material phase
 - Resistance determined by phase
- STT-MRAM
 - Inject current to change magnet polarity
 - Resistance determined by polarity
- Memristors (ReRAM)
 - Inject current to change atomic structure
 - Resistance determined by atom distance

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What is Phase Change Memory (PCM)?

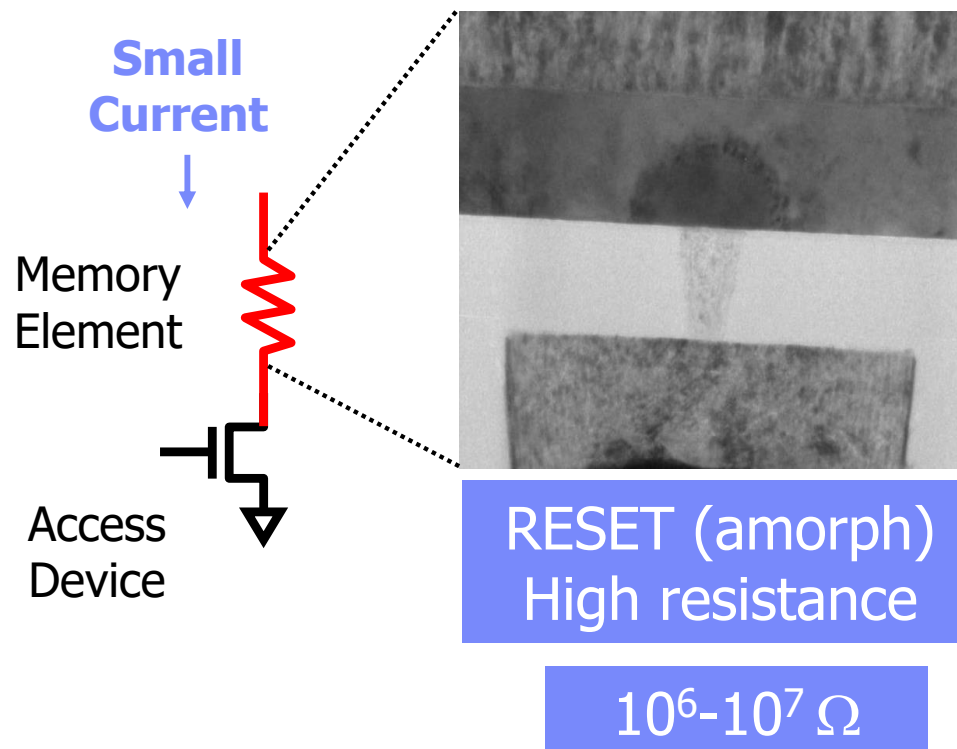
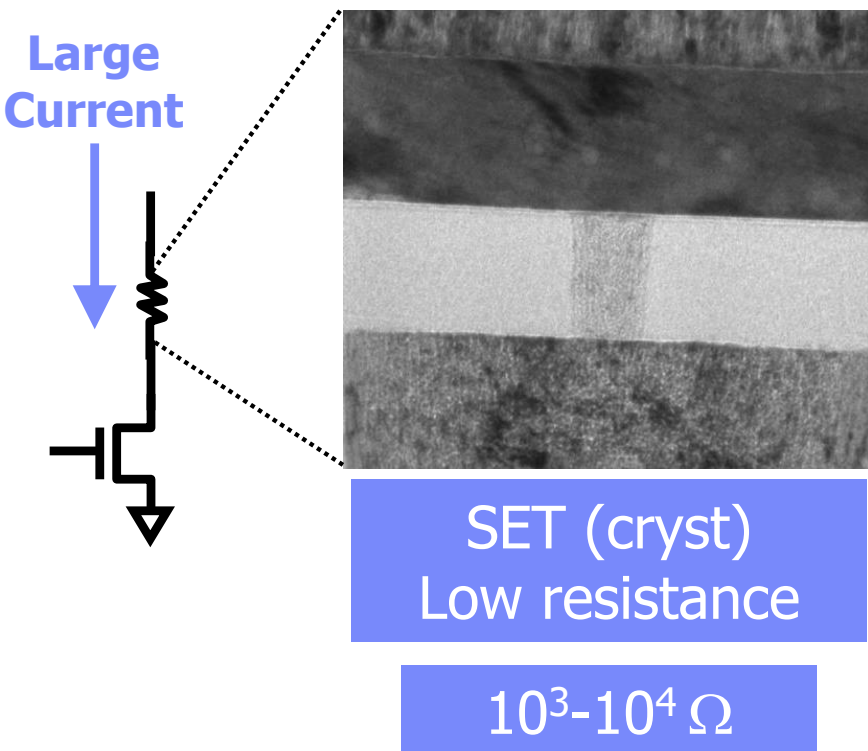
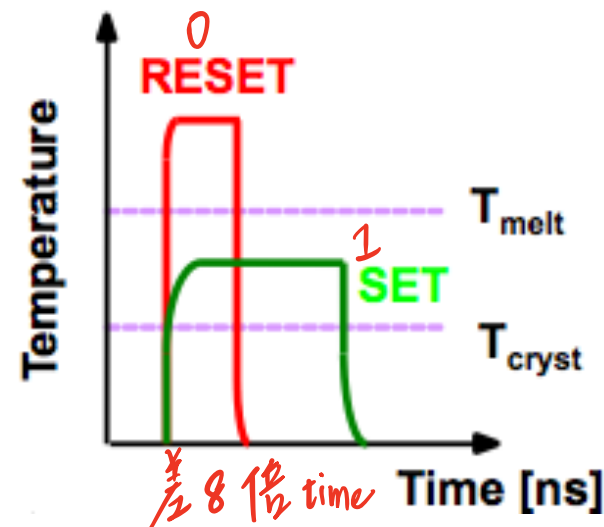
- PCM employs a reversible phase change in materials to store information
- It exploits differences in the electrical resistivity of a material in different phases
 - Amorphous: Low optical reflexivity and high electrical resistivity
 - Crystalline: High optical reflexivity and low electrical resistivity



PCM is resistive memory: High resistance (0), Low resistance (1)
 PCM cell can be switched between states reliably and quickly

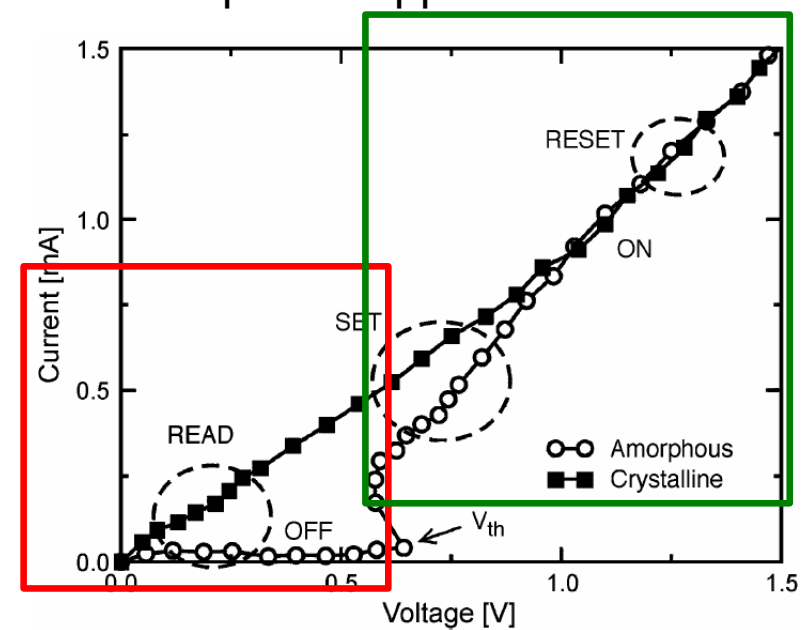
How Does PCM Work?

- Write: change phase via current injection
 - SET: sustained current to heat cell above T_{cryst}
 - RESET: cell heated above T_{melt} and quenched
- Read: detect phase via material resistance
 - amorphous/crystalline



I-V Characteristics of Set and Reset

- The set and reset states have large resistance contrast for voltage below V_{th}
 - The reset state (amorphous) is **in the high-resistance state** below V_{th}
 - Very small current ($\sim$ zero)
 - The reset state (amorphous) shows **electronic threshold switching behavior** at V_{th}
 - It is reversible if the voltage pulse is removed very quickly
 - This can be used to partial-SET the PCM cells in some specific applications
 - If the voltage is applied for longer than the crystallization time it leads to **memory switching and the cell reaches the low-resistance state** for an applied voltage larger than V_{th}



Opportunity: PCM Advantages

- Scales better than DRAM, Flash 可微縮
 - Requires current pulses, which scale linearly with feature size
 - Expected to scale to 9nm (2022 [ITRS])
 - Prototyped at 20nm (Raoux+, IBM JRD 2008)
- Can be denser than DRAM
 - Can store multiple bits per cell due to large resistance range
 - Prototypes with 2 bits/cell in ISSCC' 08, 4 bits/cell by 2012
- ★ Non-volatile
 - Retain data for >10 years at 85C
- No refresh needed, low idle power

Table 1. Technology survey.

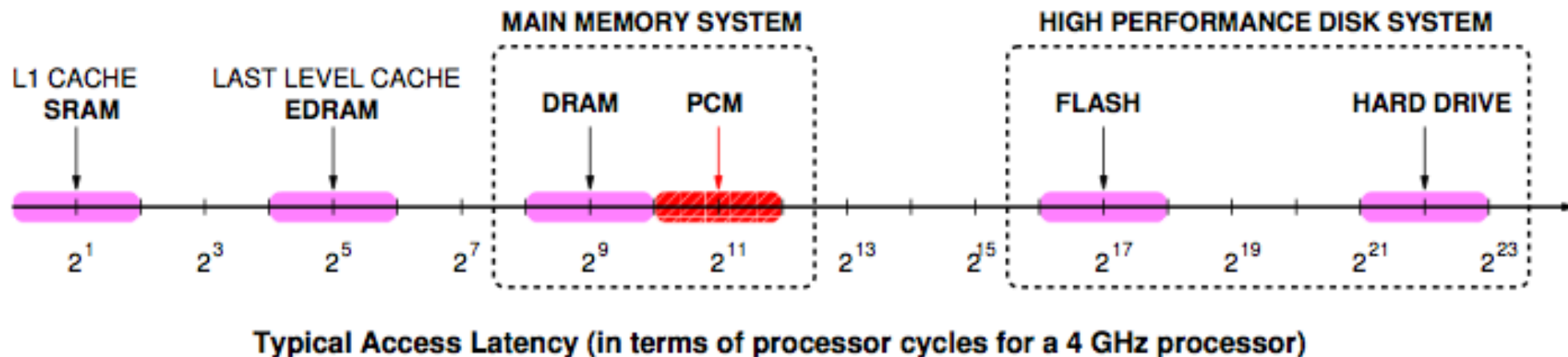
Parameter*	Published prototype									
	Horri ⁸	Ahn ¹²	Bedeschi ¹³	Oh ¹⁴	Pellizer ¹⁵	Chen ⁵	Kang ¹⁶	Bedeschi ⁹	Lee ¹⁰	Lee ²
Year	2003	2004	2004	2005	2006	2006	2006	2008	2008	**
Process, F (nm)	**	120	180	120	90	**	100	90	90	90
Array size (Mbytes)	**	64	8	64	**	**	256	256	512	**
Material	GST, N-d	GST, N-d	GST	GST	GST	GS, N-d	GST	GST	GST	GST, N-d
Cell size (μm^2)	**	0.290	0.290	**	0.097	60 nm ²	0.166	0.097	0.047	0.065 to 0.097
Cell size, F^2	**	20.1	9.0	**	12.0	**	16.6	12.0	5.8	9.0 to 12.0
Access device	**	**	BJT	FET	BJT	**	FET	BJT	Diode	BJT
Read time (ns)	**	70	48	68	**	**	62	**	55	48
Read current (μA)	**	**	40	**	**	**	**	**	**	40
Read voltage (V)	**	3.0	1.0	1.8	1.6	**	1.8	**	1.8	1.0
Read power (μW)	**	**	40	**	**	**	**	**	**	40
Read energy (pJ)	**	**	2.0	**	**	**	**	**	**	2.0
Set time (ns)	100	150	150	180	**	80	300	**	400	150
Set current (μA)	200	**	300	200	**	55	**	**	**	150
Set voltage (V)	**	**	2.0	**	**	1.25	**	**	**	1.2
Set power (μW)	**	**	300	**	**	34.4	**	**	**	90
Set energy (pJ)	**	**	45	**	**	2.8	**	**	**	13.5
Reset time (ns)	50	10	40	10	**	60	50	**	50	40
Reset current (μA)	600	600	600	600	400	90	600	300	600	300
Reset voltage (V)	**	**	2.7	**	1.8	1.6	**	1.6	**	1.6
Reset power (μW)	**	**	1620	**	**	80.4	**	**	**	480
Reset energy (pJ)	**	**	64.8	**	**	4.8	**	**	**	19.2
Write endurance (MLC)	10^7	10^9	10^6	**	10^8	10^4	**	10^5	10^5	10^8

* BJT: bipolar junction transistor; FET: field-effect transistor; GST: $\text{Ge}_2\text{Sb}_2\text{Te}_5$; MLC: multilevel cells; N-d: nitrogen doped.

** This information is not available in the publication cited.

Phase Change Memory Properties: Latency

- Latency comparable to, but slower than DRAM



- Read Latency
 - 50ns: 4x DRAM, 10^{-3} x NAND Flash
- Write Latency
 - 150ns: 12x DRAM
- Write Bandwidth
 - 5-10 MB/s: 0.1x DRAM, 1x NAND Flash

Phase Change Memory Properties

- Dynamic Energy

- 40 μ A Rd, 150 μ A Wr
- 2-43x DRAM, 1x NAND Flash

- Endurance

- Writes induce phase change at 650C
- Contacts degrade from thermal expansion/contraction
- 10^8 writes per cell $\Rightarrow$ 讀寫頭易壞
- 10^{-8} x DRAM, 10^3 x NAND Flash

- Cell Size

- 9-12F² using BJT, single-level cells
- 1.5x DRAM, 2-3x NAND (will scale with feature size, MLC)

Technology Attributes

R. Bez, LETI Workshop, 2009

Attributes	PCM	DRAM	NAND	NOR
Scaling	Longest Term	32nm	25nm	32nm
Bit Alterable	Yes	Yes	No	No
Byte Addressable	Yes	Yes	No	Yes
Non-Volatile	Yes	No	Yes	Yes
Random Read	20~50ns	20ns	25000ns	100ns
Write Speed	90ns (0) 120ns (1)	20ns	>10MB/s	1MB/s
Power	Low	High	Low	Low
Energy Write	6 J/GB	1 J/GB	17.5 J/GB	
Energy Read	1 J/GB	0.8 J/GB	1.5 J/GB	
Endurance	$10^6 \sim 10^8$	10^{15}	$10^3 \sim 10^5$	

Shimin Chen, Phillip B. Gibbons, Suman Nath, "Rethinking Database Algorithms for Phase Change Memory," Biennial Conference on Innovative Data Systems Research (CIDR), 2011.

Technology Attributes (Cont.)

Media	Access time		
	Read	Write	Erase
DRAM	20ns (1B) 10us (512B)	20ns (1B) 10us (512B)	N/A
NOR Flash	45ns (1B) 23.0us (512B)	14us (1B) 7.2ms (512B)	18ms (128KB)
PCM	20 - 50 ns (1B)	90-120 ns (1B)	N/A
NAND Flash (SLC)	15us (1B) MAX: 35us (8KB)	300us (1B) TYP: 350us (8KB) MAX: 500us (8KB)	TYP: 1.5ms (1MB) MAX: 3ms (1MB)
DISK	15.8ms TYP: 8.2ms (512B)	6.06ms TYP: 9.2ms (512B)	N/A

Reference Devices/Modules: DRAM: Micron DDR3-1333. NOR Flash: Silicon Storage Technology SST39LF020. PCM: Micron P8P Parallel PCM, NAND Flash: Micron MT29F256G08AUAAC5. Disk: Deskstar™ 7K3000 series.

Phase Change Memory: Pros and Cons

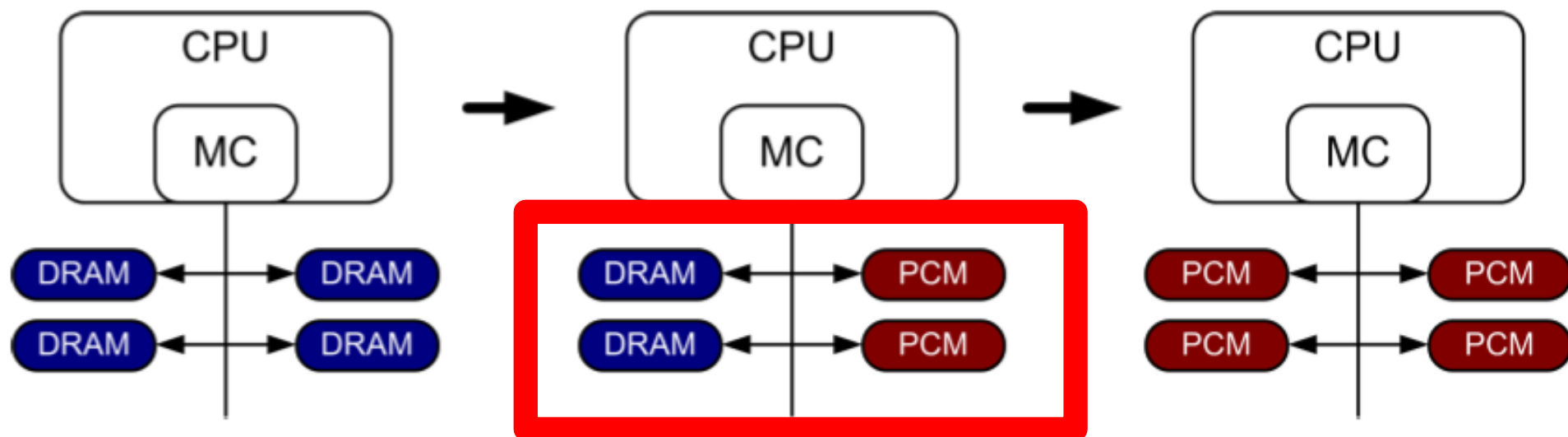
- Pros over DRAM
 - Better technology scaling (capacity and cost)
 - Non volatility > Persistent
 - Low idle power (no refresh)
- Cons
 - Higher latencies: ~4-15x DRAM (especially write)
 - Higher active energy: ~2-50x DRAM (especially write)
 - Lower endurance (a cell dies after $\sim 10^8$ writes)
 - Reliability issues (resistant drift)
- Challenges in enabling PCM as DRAM replacement/helper:
 - Mitigate PCM shortcomings
 - Find the right way to place PCM in the system
 - Ensure secure and fault-tolerant PCM operation

PCM-based Main Memory: Research Challenges

- Where to place PCM in the memory hierarchy?
 - Hybrid OS controlled PCM-DRAM
 - Hybrid OS controlled PCM and hardware-controlled DRAM
 - Pure PCM main memory
- How to mitigate shortcomings of PCM?
- How to minimize amount of DRAM in the system?
- How to take advantage of (byte-addressable and fast) non-volatile main memory?
- Can we design specific-NVM-technology-agnostic techniques?

PCM-based Main Memory (I)

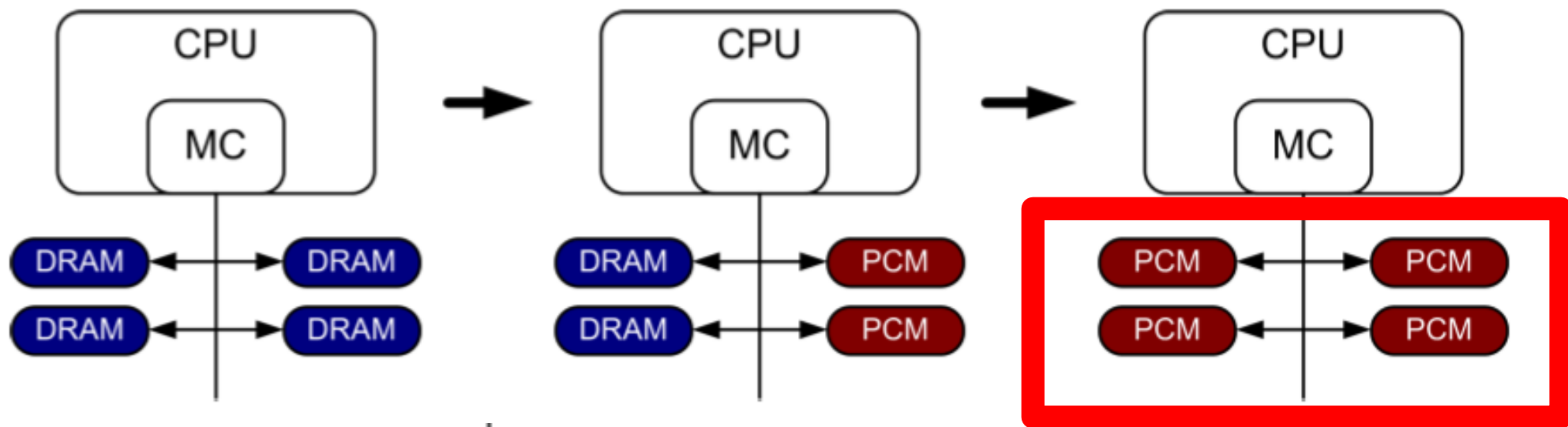
- How should PCM-based (main) memory be organized?



- **Hybrid PCM+DRAM** [Qureshi+ ISCA'09, Dhiman+ DAC'09, Meza+ IEEE CAL'12]:
 - How to partition/migrate data between PCM and DRAM

PCM-based Main Memory (II)

- How should PCM-based (main) memory be organized?



- **Pure PCM main memory** [Lee et al., ISCA'09, Top Picks'10]:
 - How to redesign entire hierarchy (and cores) to overcome PCM shortcomings

Hybrid Memory Systems: Challenges

- Partitioning
 - Should DRAM be a cache or main memory, or configurable?
 - What fraction? How many controllers?
- Data allocation/movement (energy, performance, lifetime)
 - Who manages allocation/movement?
 - What are good control algorithms?
 - How do we prevent degradation of service due to wearout?
- Design of cache hierarchy, memory controllers, OS
 - Mitigate PCM shortcomings, exploit PCM advantages
- Design of PCM/DRAM chips and modules
 - Rethink the design of PCM/DRAM with new requirements

Agenda

- Overview on the main memory system
- Opportunity: Emerging Memory Technologies
 - Background
 - Phase Change Memory
 - **PCM (or Technology X) as DRAM Replacement**
 - Hybrid Memory Systems
- Conclusion

An Initial Study: Replace DRAM with PCM

- Lee, Ipek, Mutlu, Burger, “[Architecting Phase Change Memory as a Scalable DRAM Alternative](#),” ISCA 2009.
 - Surveyed prototypes from 2003-2008 (e.g. IEDM, VLSI, ISSCC)
 - Derived “average” PCM parameters for F=90nm

Density

- ▶ $9 - 12F^2$ using BJT
- ▶ $1.5\times$ DRAM

Latency

- ▶ 50ns Rd, 150ns Wr
- ▶ $4\times, 12\times$ DRAM

Endurance

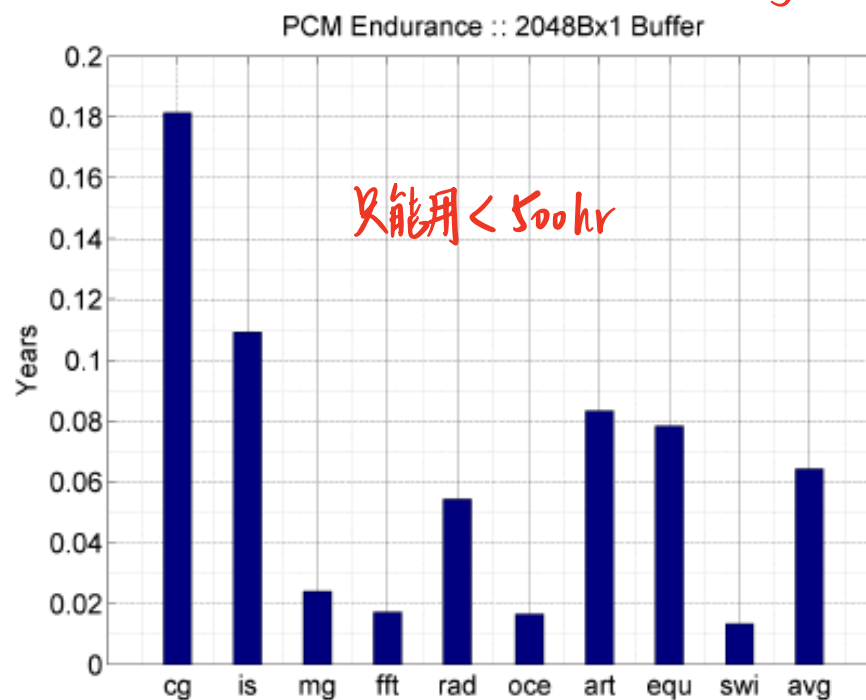
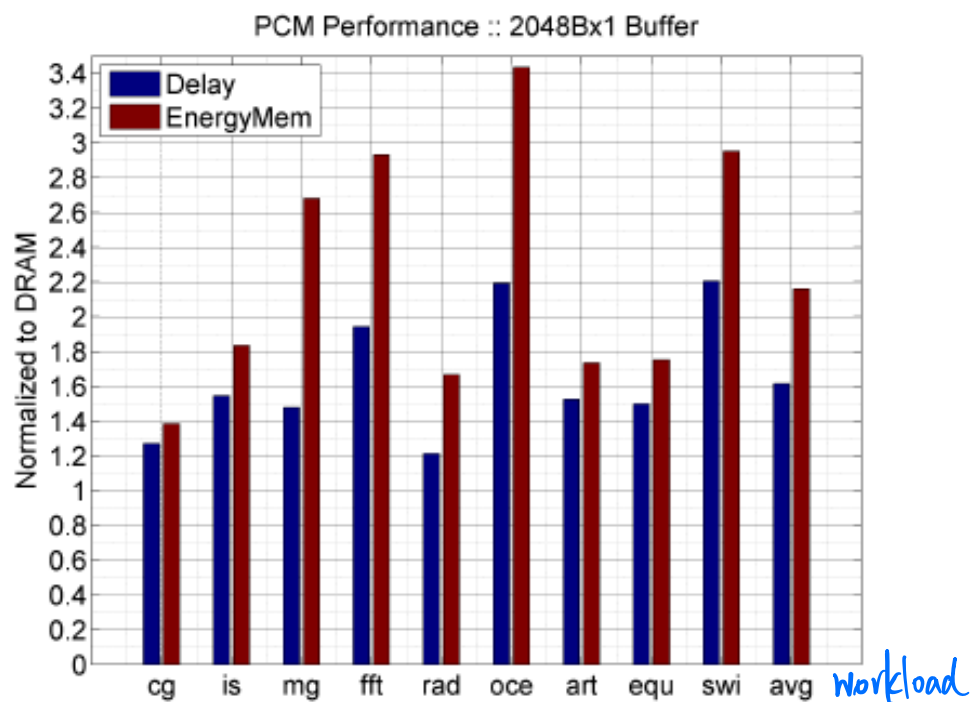
- ▶ $1E+08$ writes
- ▶ $1E-08\times$ DRAM

Energy

- ▶ $40\mu A$ Rd, $150\mu A$ Wr
- ▶ $2\times, 43\times$ DRAM

Results: Naïve Replacement of DRAM with PCM

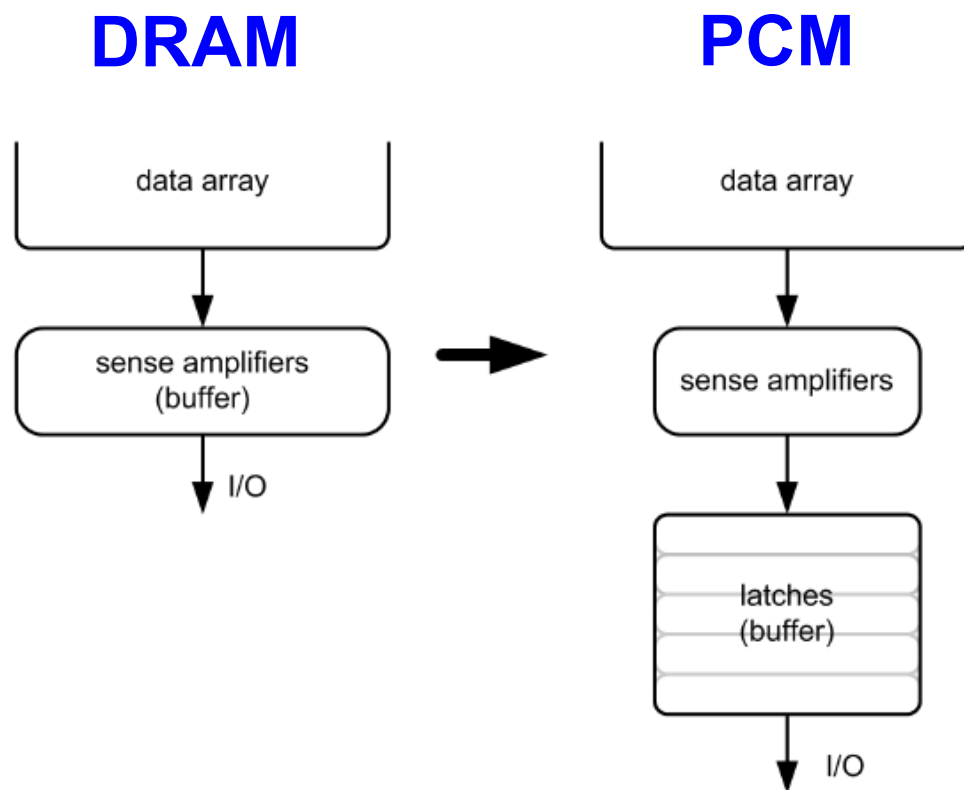
- Replace DRAM with PCM in a 4-core, 4MB L2 system
- PCM organized the same as DRAM: row buffers, banks, peripherals
- 1.6x delay, 2.2x energy, 500-hour average lifetime $\Rightarrow$ 比DRAM慢. 耗能



- Lee, Ipek, Mutlu, Burger, “Architecting Phase Change Memory as a Scalable DRAM Alternative,” ISCA 2009.

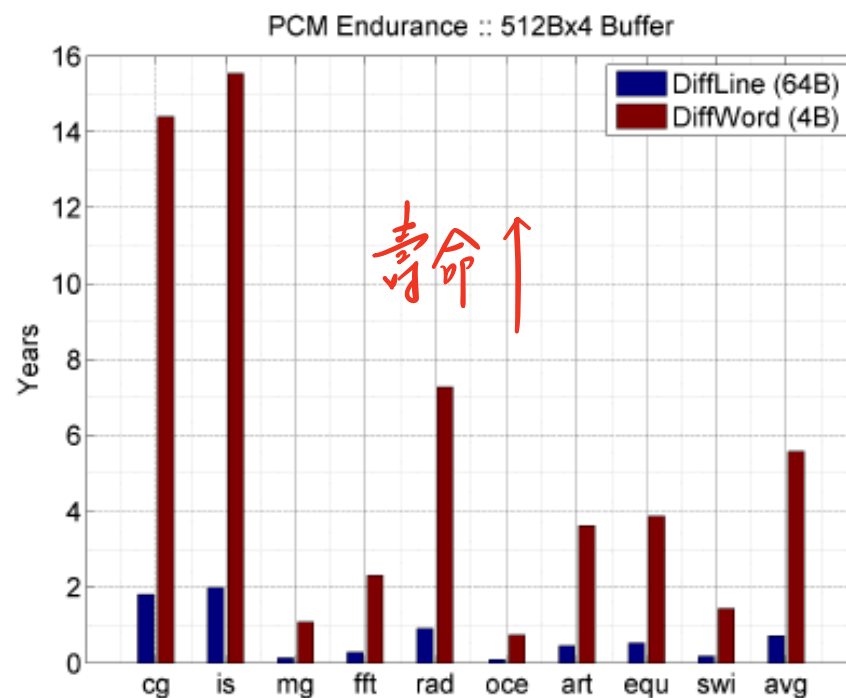
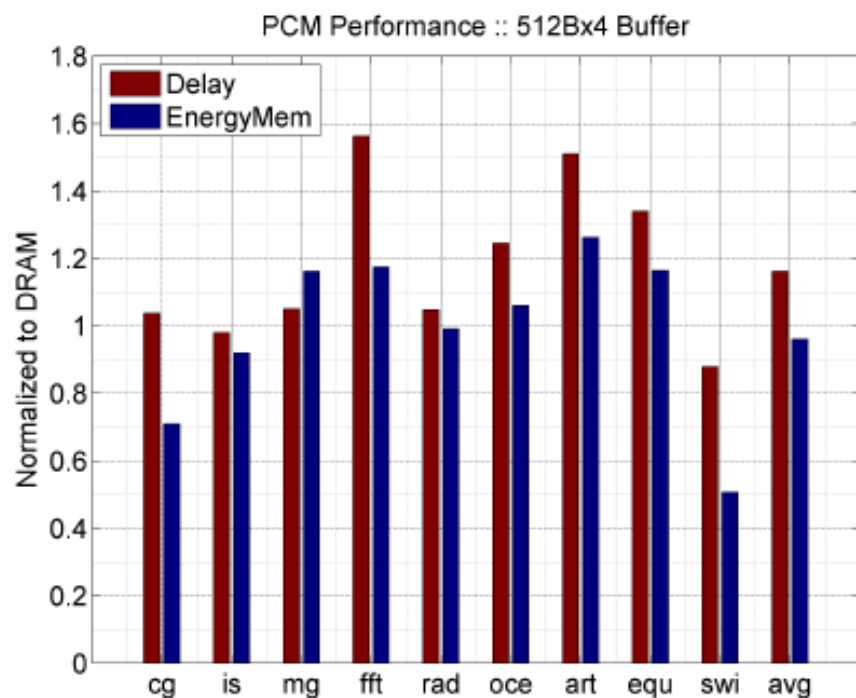
Architecting PCM to Mitigate Shortcomings

- Idea 1: Use multiple narrow row buffers in each PCM chip
 - Reduces array reads/writes → better endurance, latency, energy
 - Using smaller row buffers but many of them
 - Total amount of buffer is still the same
- Idea 2: Write into array at cache block or word granularity
 - Only updates the necessary write
 - Reduces unnecessary wear



Results: Architected PCM as Main Memory

- 1.2x delay, 1.0x energy, 5.6-year average lifetime $\Rightarrow$ 和 DRAM 差不多了
- Scaling improves energy, endurance, density



- Caveat 1: Worst-case lifetime is much shorter (no guarantees)
- Caveat 2: Intensive applications see large performance and energy hits
- Caveat 3: Optimistic PCM parameters?

Intel Optane Memory (Idea Realized in 2019)

- Non-volatile main memory
- Based on 3D-XPoint Technology



Some PCM Bits Take Longer to Read...

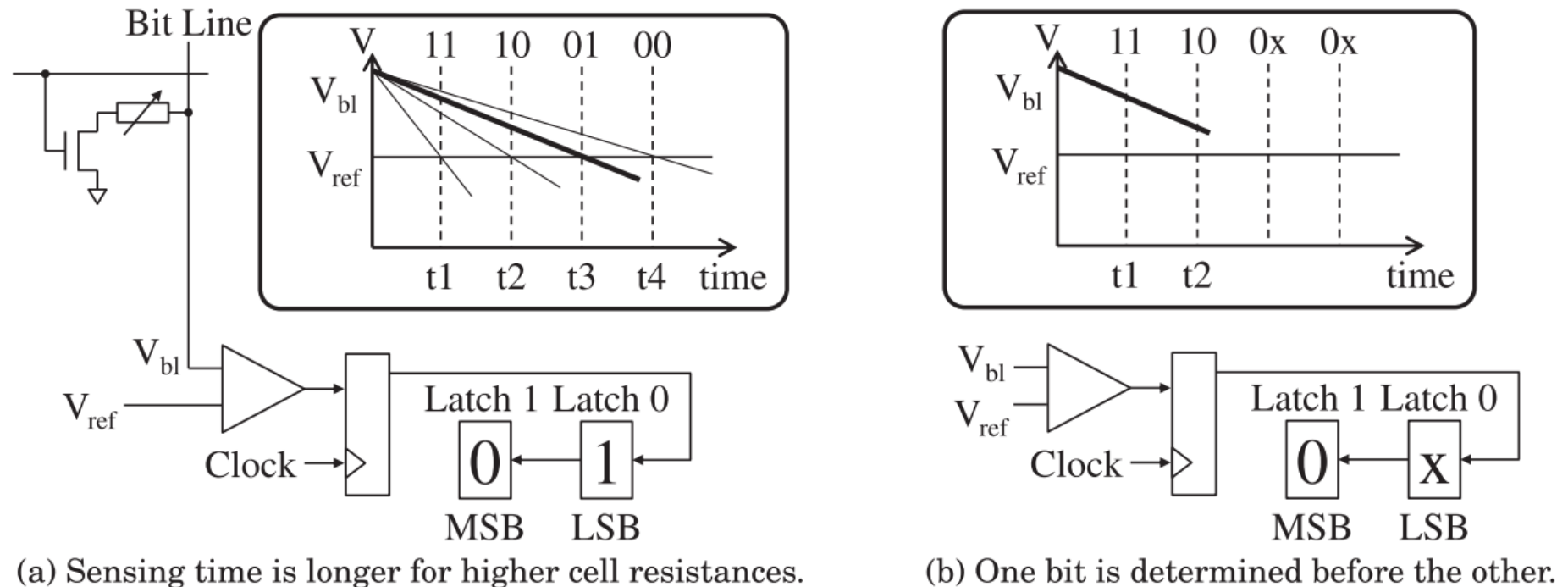


Fig. 3. MLC PCM cell read operation [Qureshi et al. 2010b].

Efficient Data Mapping and Buffering Techniques for Multilevel Cell Phase-Change Memories

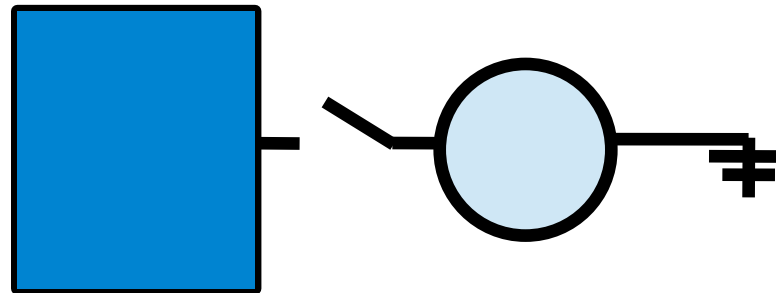
HANBIN YOON* and JUSTIN MEZA, Carnegie Mellon University
 NAVEEN MURALIMANOVAR, Hewlett-Packard Labs
 NORMAN P. JOUPPI**, Google Inc.
 ONUR MUTLU, Carnegie Mellon University

Observation 1: Read Asymmetry

- *The read latency/energy of Bit 1^{MSB} is lower than that of Bit 0^{LSB}*
- This is due to how MLC-PCM cells are read

Observation 1: Read Asymmetry

Simplified example



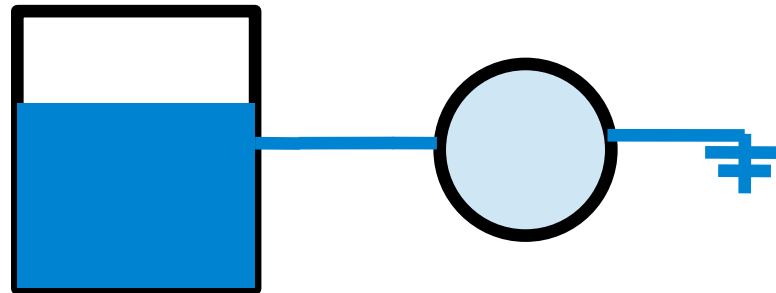
Capacitor filled
with reference
voltage

MLC-PCM cell
with unknown
resistance



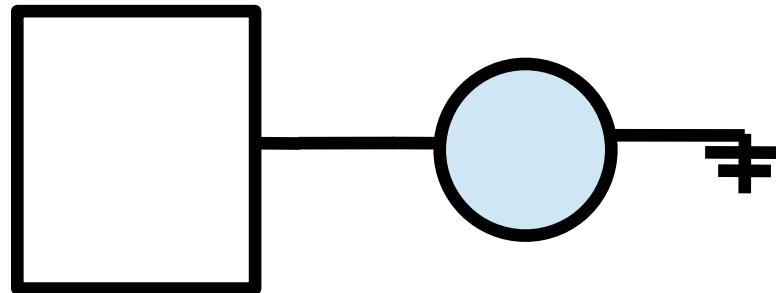
Observation 1: Read Asymmetry

Simplified example



Observation 1: Read Asymmetry

Simplified example

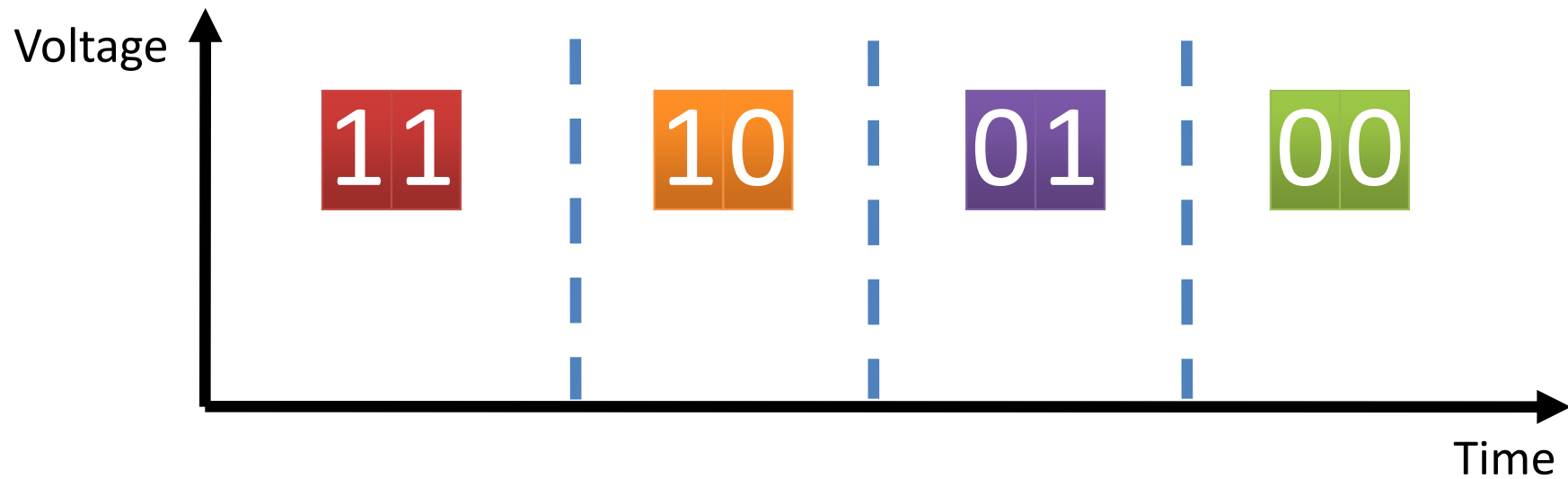


Infer data value

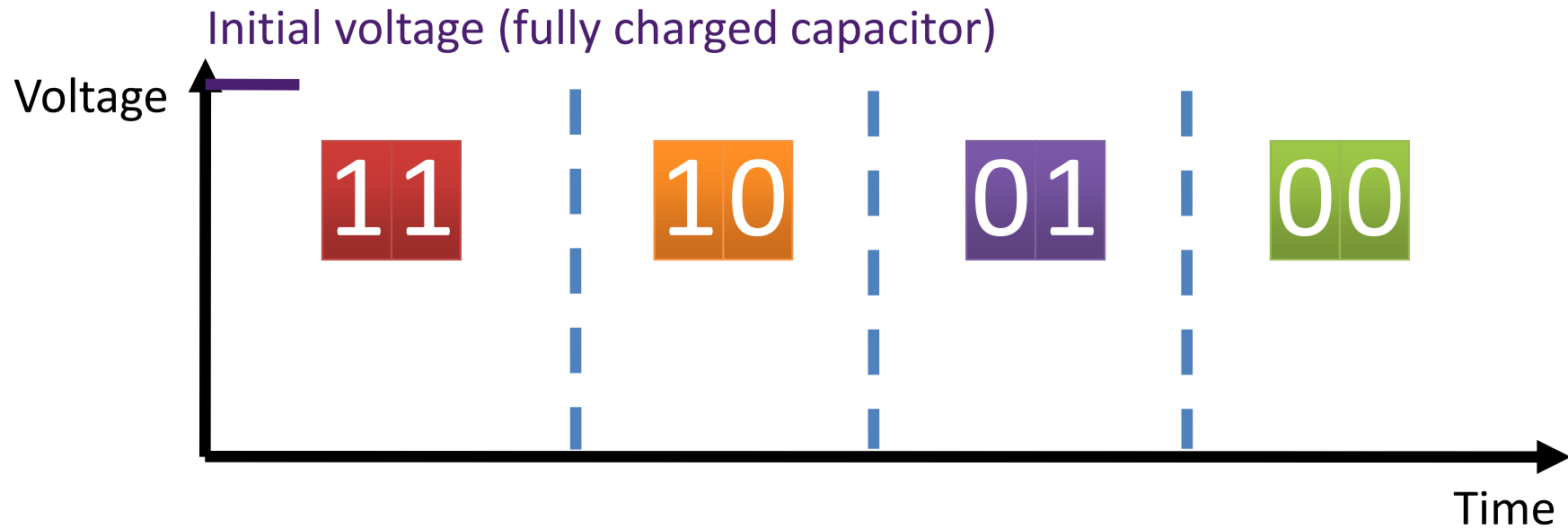
Observation 1: Read Asymmetry



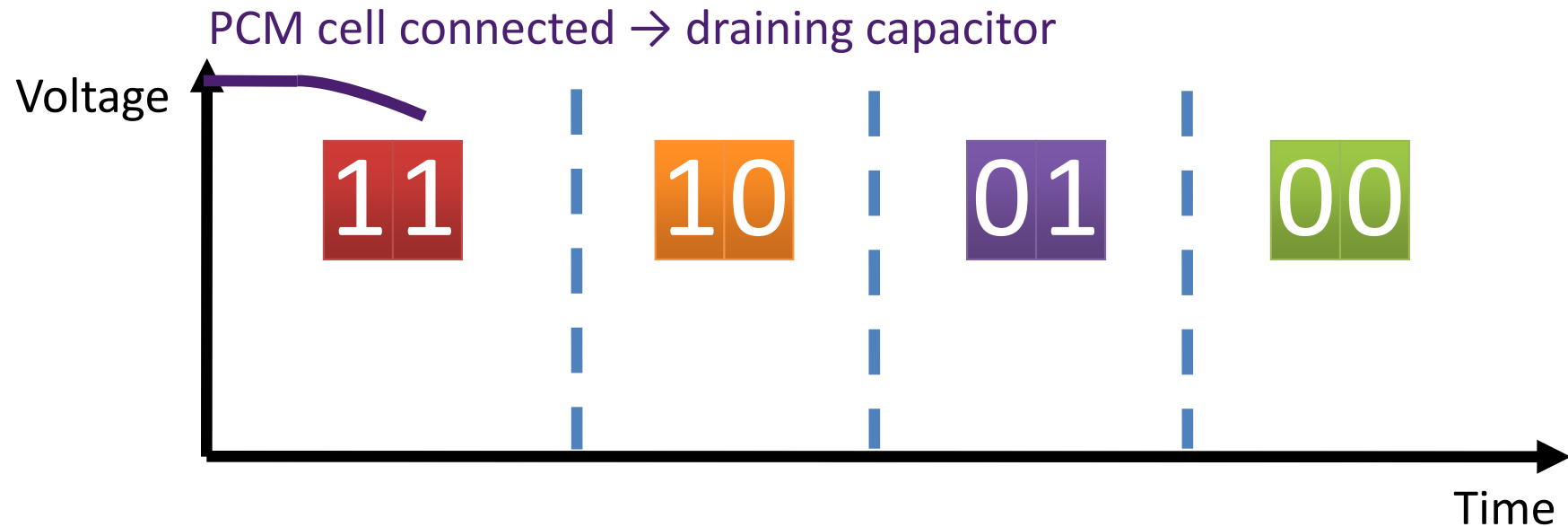
Observation 1: Read Asymmetry



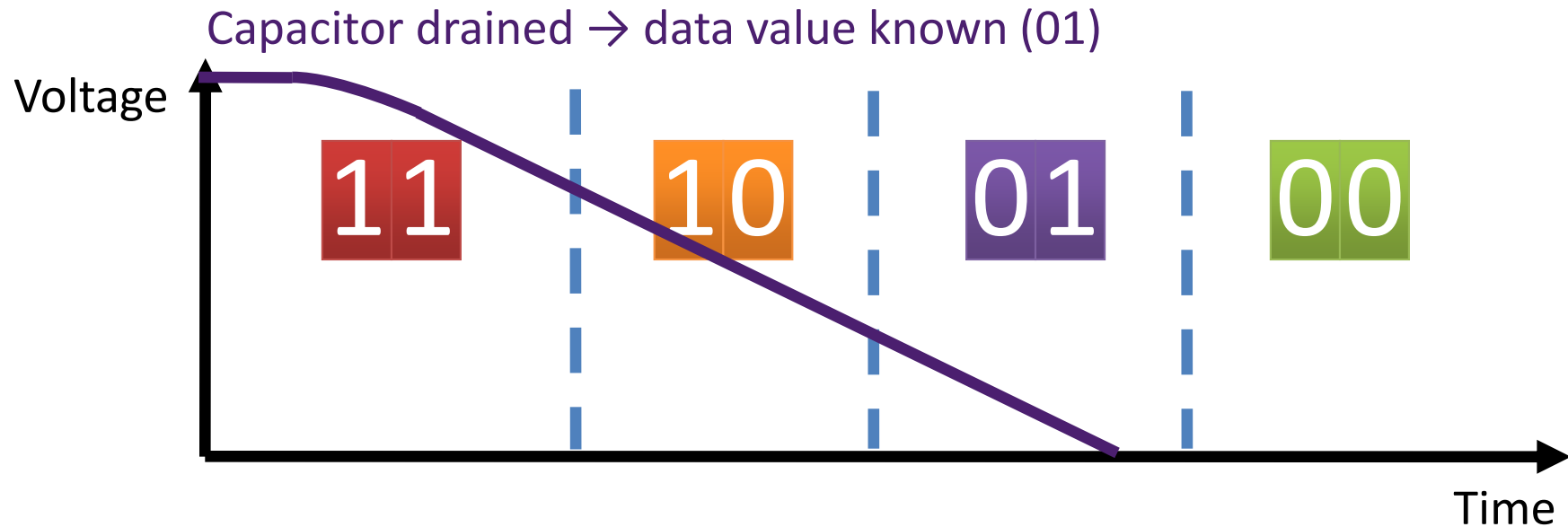
Observation 1: Read Asymmetry



Observation 1: Read Asymmetry



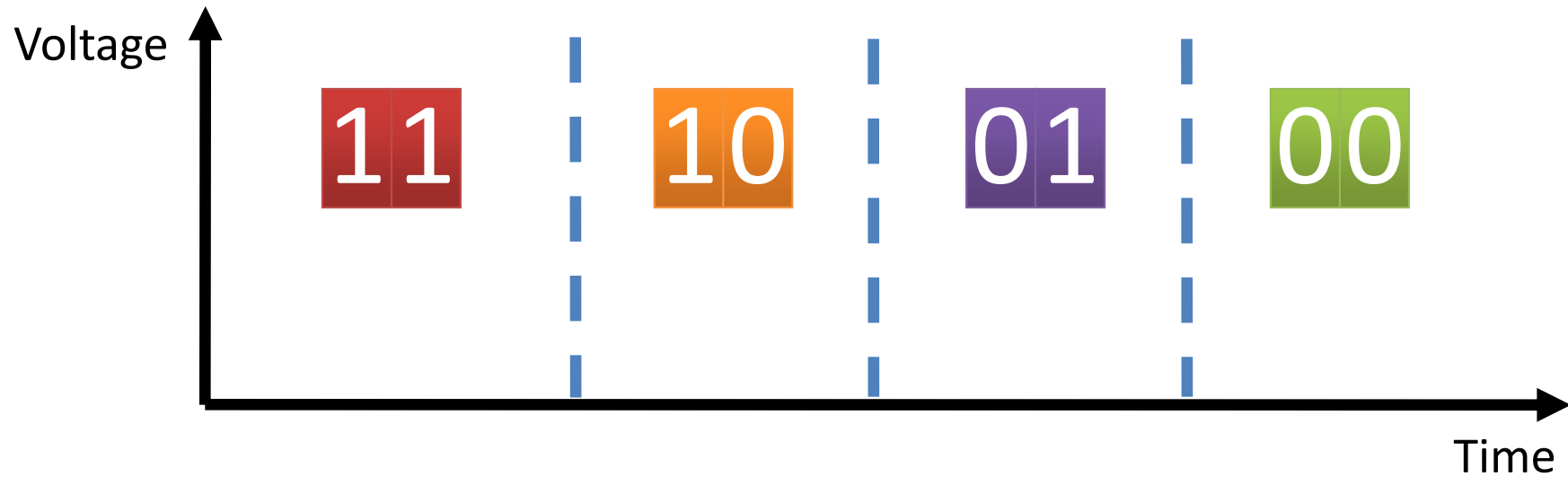
Observation 1: Read Asymmetry



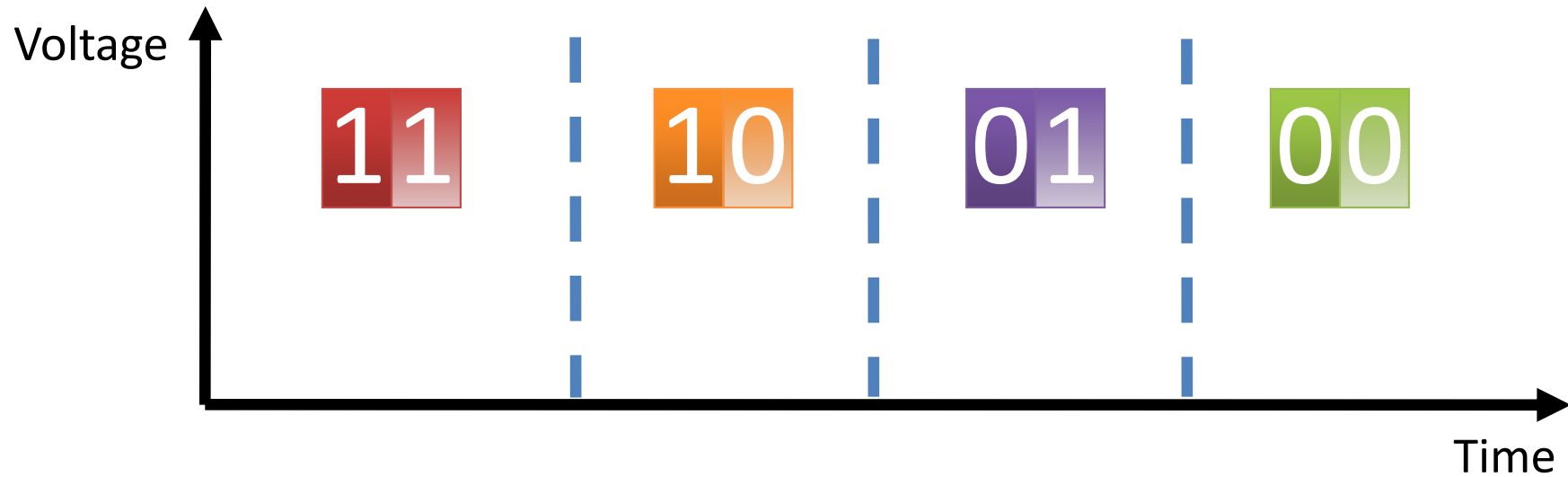
Observation 1: Read Asymmetry

- In existing devices
 - Both MLC bits are read at the same time
 - Must wait *maximum time* to read both bits
- However, *we can infer information about Bit 1 before this time*

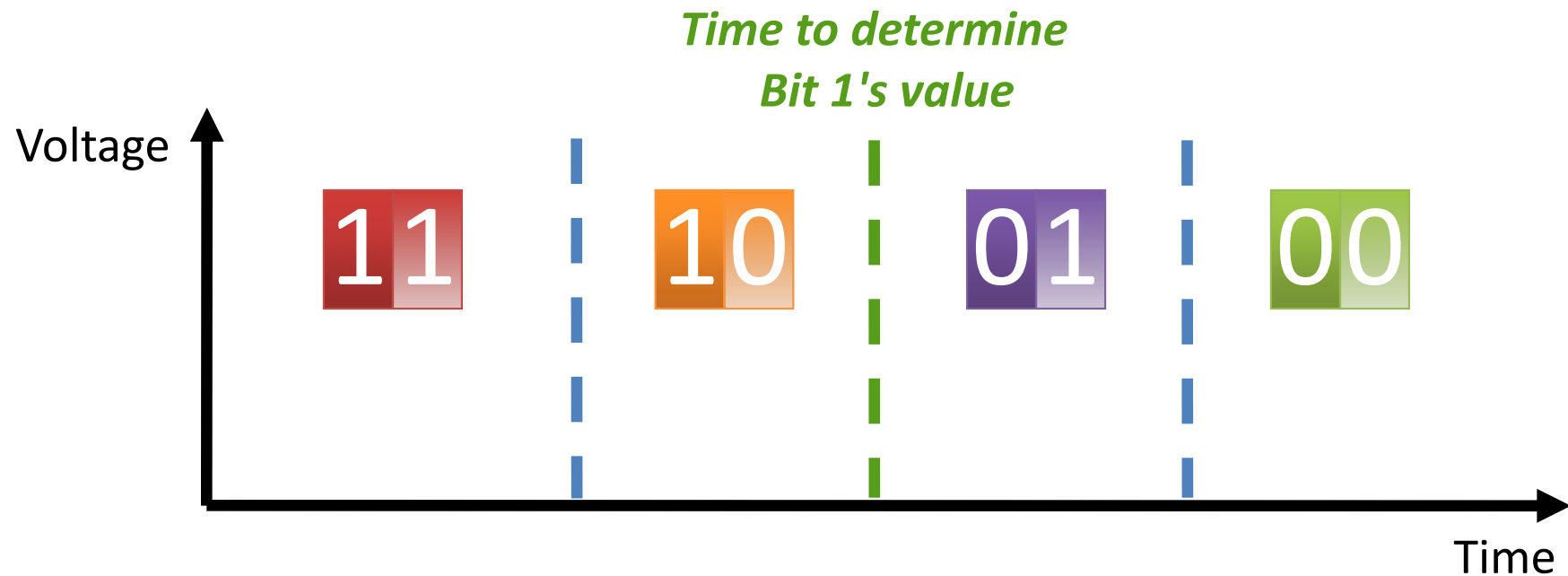
Observation 1: Read Asymmetry



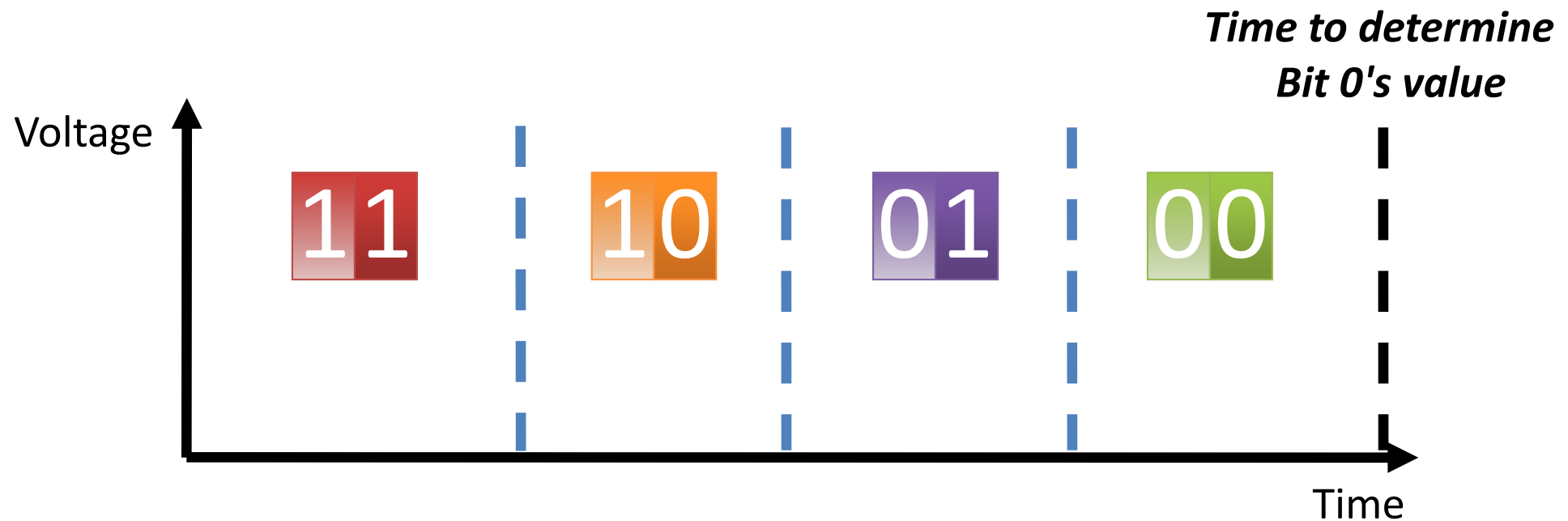
Observation 1: Read Asymmetry



Observation 1: Read Asymmetry



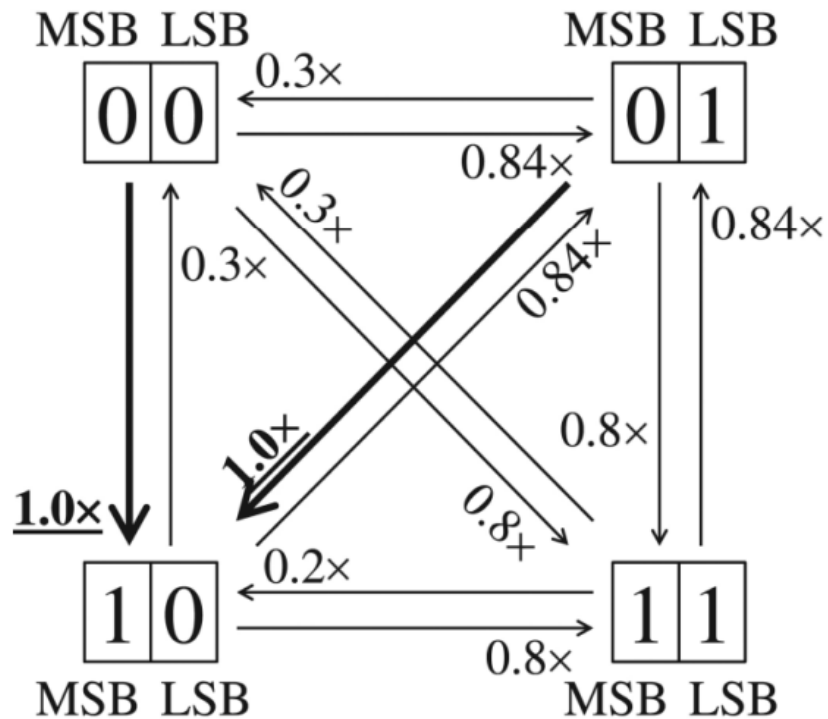
Observation 1: Read Asymmetry



Some PCM Bits Take Longer to Write...

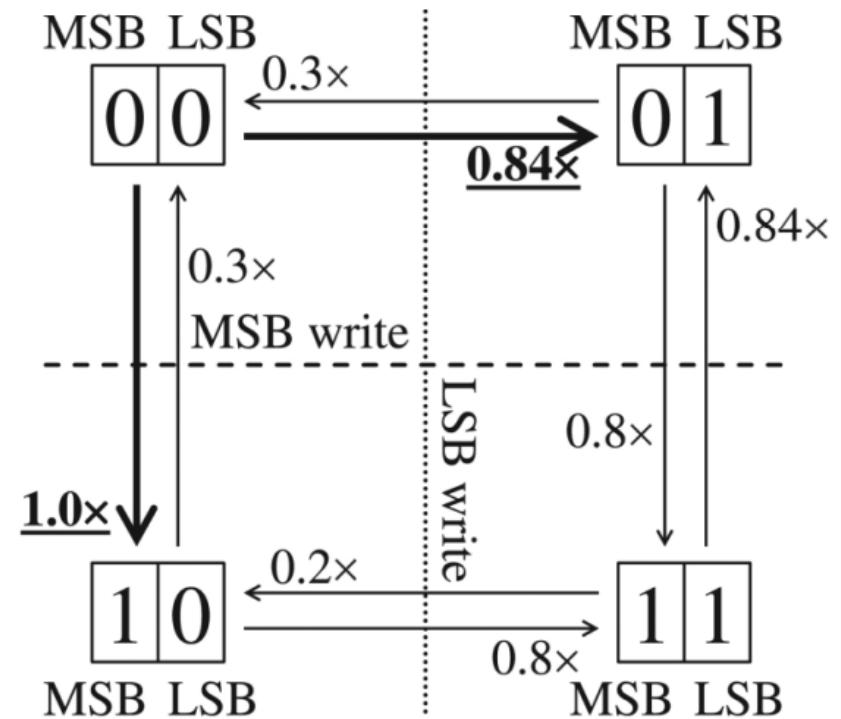
Efficient Data Mapping and Buffering Techniques for MLC PCM

40:7



(a) All possible cell state transitions.

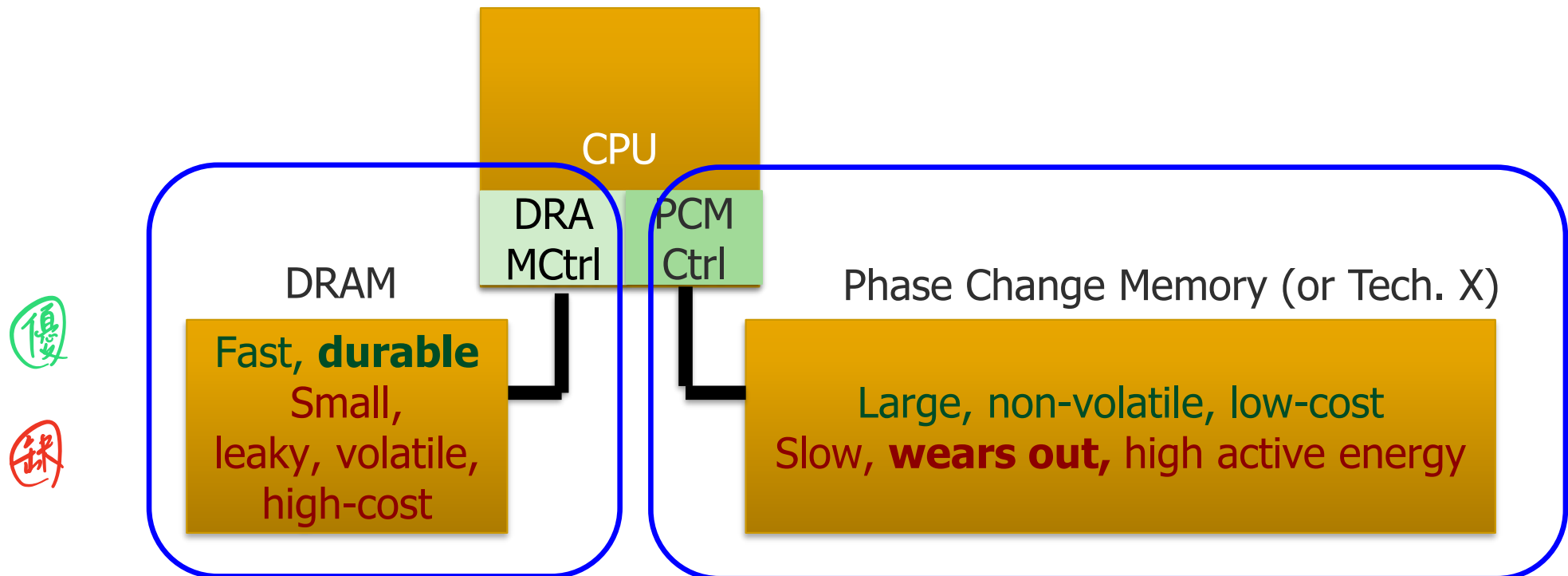
↪ 0 變成 1 較慢



(b) Cell state transitions when modifying only the MSB or the LSB.

Fig. 4. MLC PCM cell write latencies [Joshi et al. 2011; Nirschl et al. 2007; Happ et al. 2006].

Hybrid Memory Systems



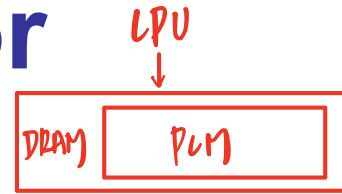
Hardware/software manage data allocation and movement
to achieve the best of multiple technologies

Meza+, "Enabling Efficient and Scalable Hybrid Memories," IEEE Comp. Arch. Letters, 2012.
Yoon, Meza et al., "Row Buffer Locality Aware Caching Policies for Hybrid Memories," ICCD 2012 Best Paper Award.

Hybrid Memory Systems: Issues

- Cache vs. Main Memory
- ^快 Granularity of Data Move/Management: Fine or Coarse
- Hardware vs. Software vs. HW/SW Cooperative
- When to migrate data?
- How to design a scalable and efficient large cache?
- ...

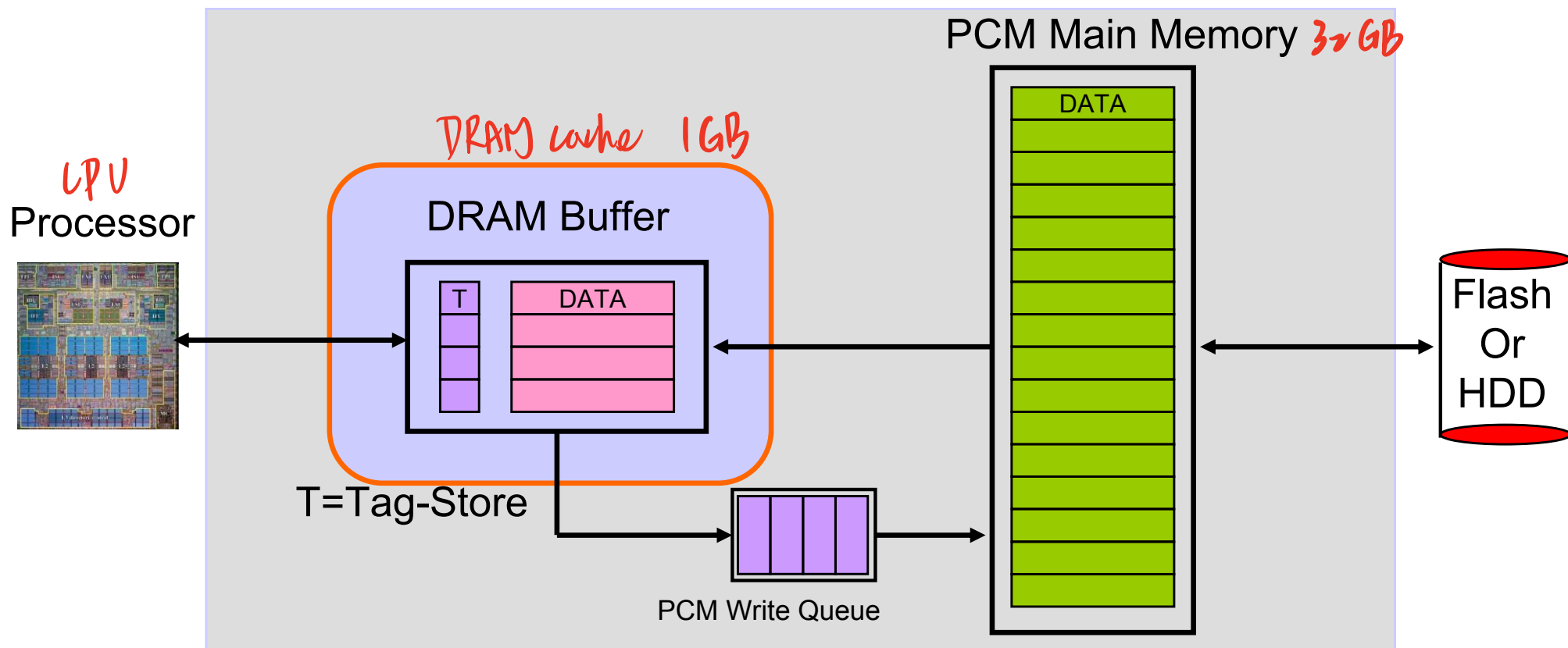
One Option: DRAM as a Cache for PCM



- PCM is main memory; DRAM caches memory rows/blocks
 - Benefits: Reduced latency on DRAM cache hit; write filtering
- Memory controller hardware manages the DRAM cache
 - Benefit: Eliminates system software overhead
- Three issues:
 - What data should be placed in DRAM versus kept in PCM?
 - What is the granularity of data movement?
 - How to design a low-cost hardware-managed DRAM cache?
- Two idea directions:
 - Locality-aware data placement [Yoon+, ICCD 2012]
 - Cheap tag stores and dynamic granularity [Meza+, IEEE CAL 2012]

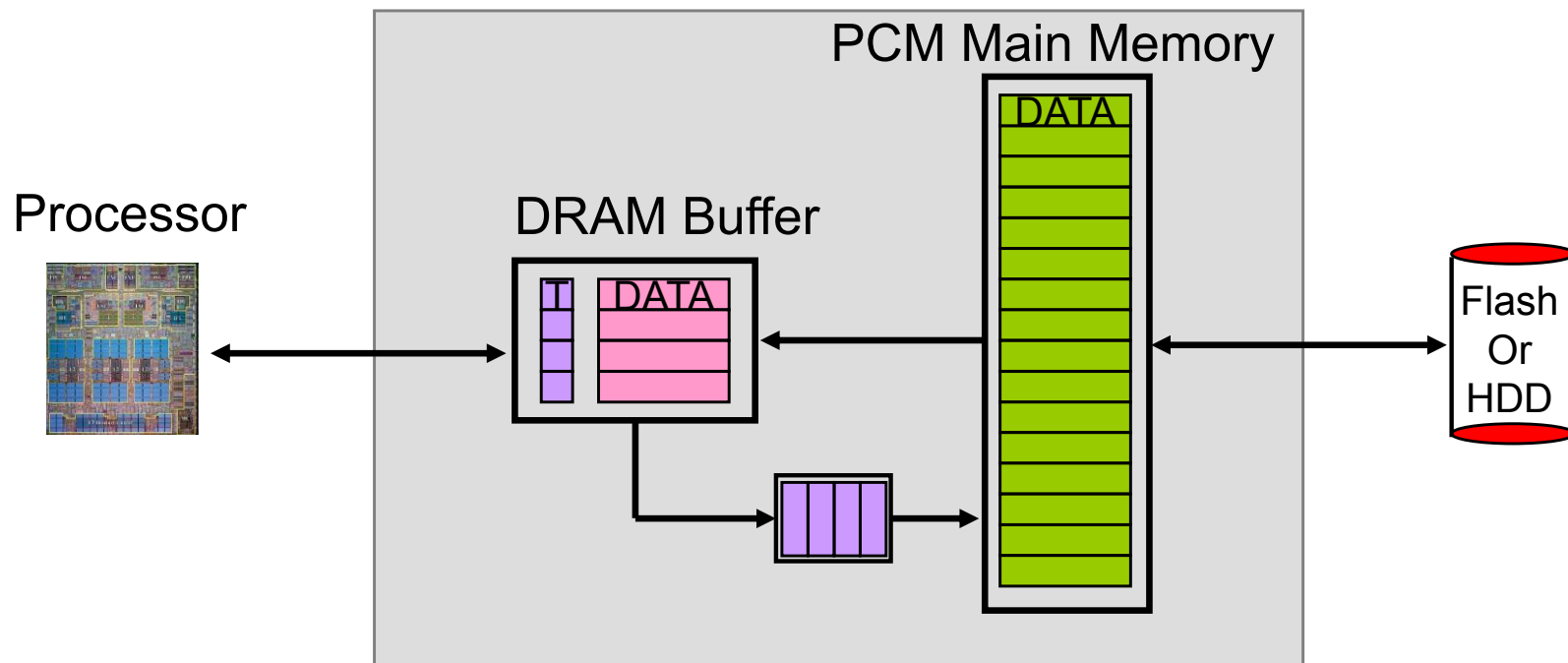
DRAM as a Cache for PCM

- Goal: Achieve the best of both DRAM and PCM/NVM
 - Minimize amount of DRAM w/o sacrificing performance, endurance
 - DRAM as cache to tolerate PCM latency and write bandwidth
 - PCM as main memory to provide large capacity at good cost and power



Write Filtering Techniques

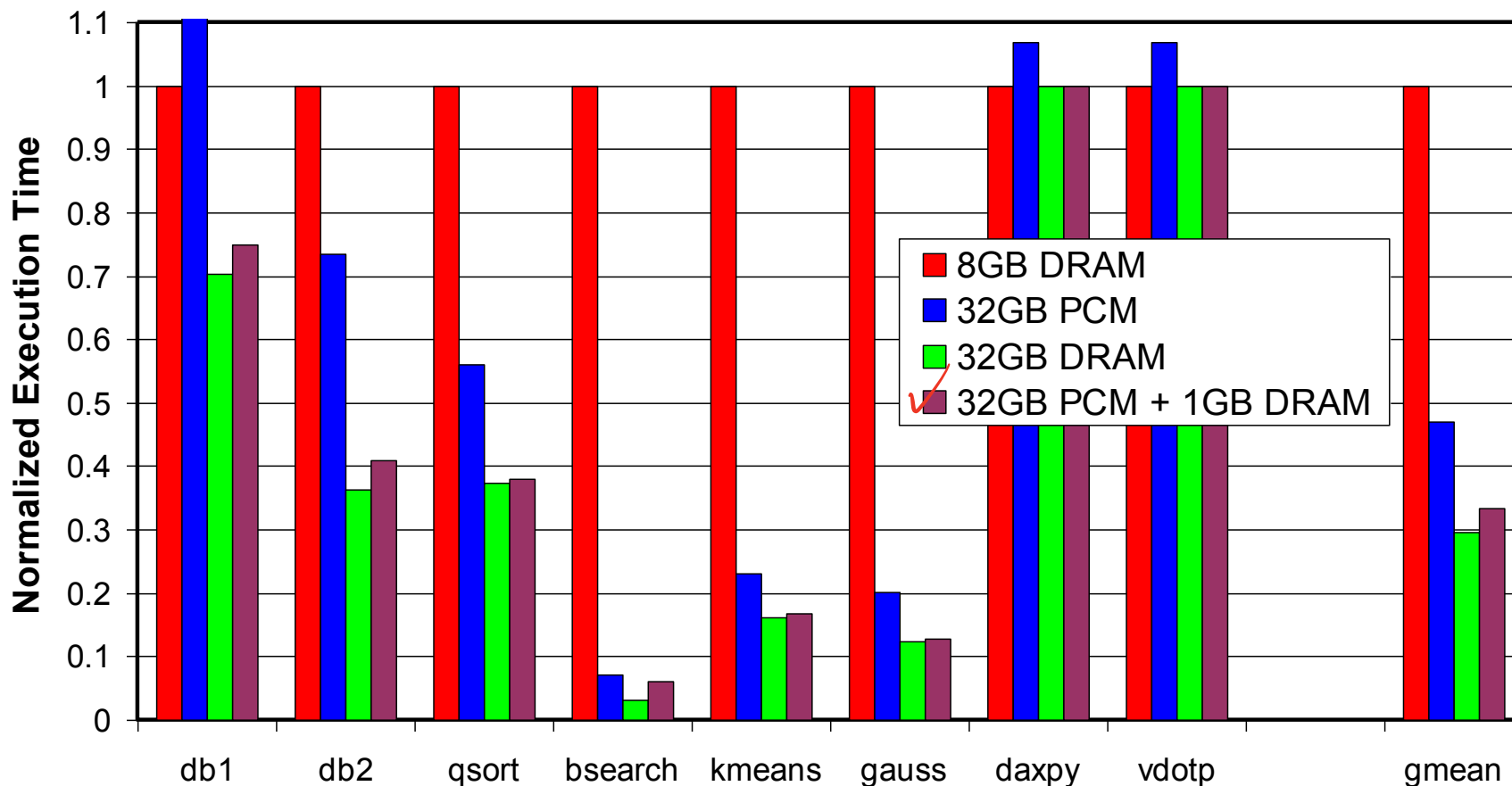
- Lazy Write: Pages from disk installed only in DRAM, not PCM
- Partial Writes: Only dirty lines from DRAM page written back
- Page Bypass: Discard pages with poor reuse on DRAM eviction



- Qureshi et al., “**Scalable high performance main memory system using phase-change memory technology**,” ISCA 2009.

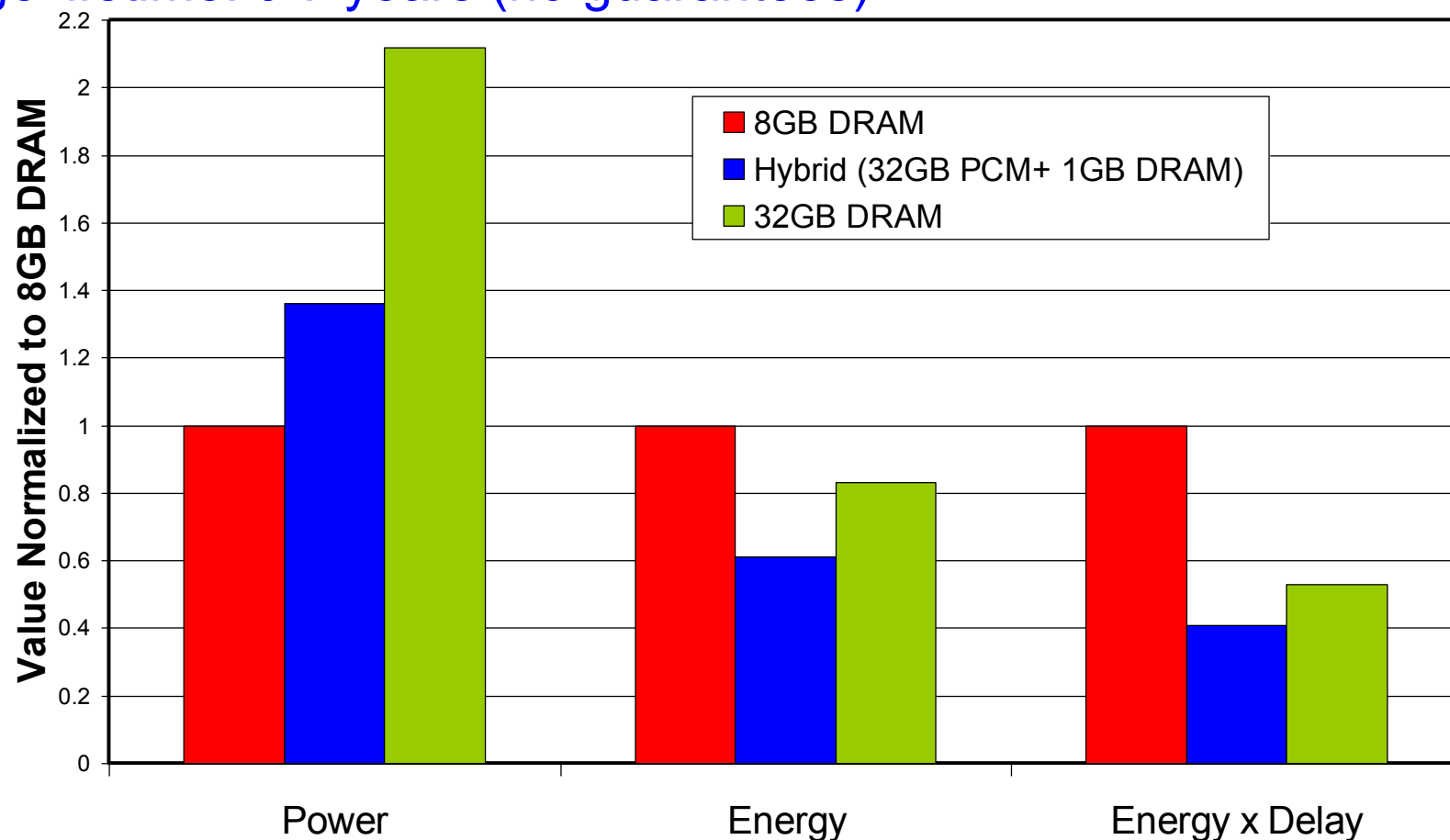
Results: DRAM as PCM Cache (I)

- Simulation of 16-core system, 8GB DRAM main-memory at 320 cycles, HDD (2 ms) with Flash (32 us) with Flash hit-rate of 99%
- Assumption: PCM 4x denser, 4x slower than DRAM
- DRAM block size = PCM page size (4kB)



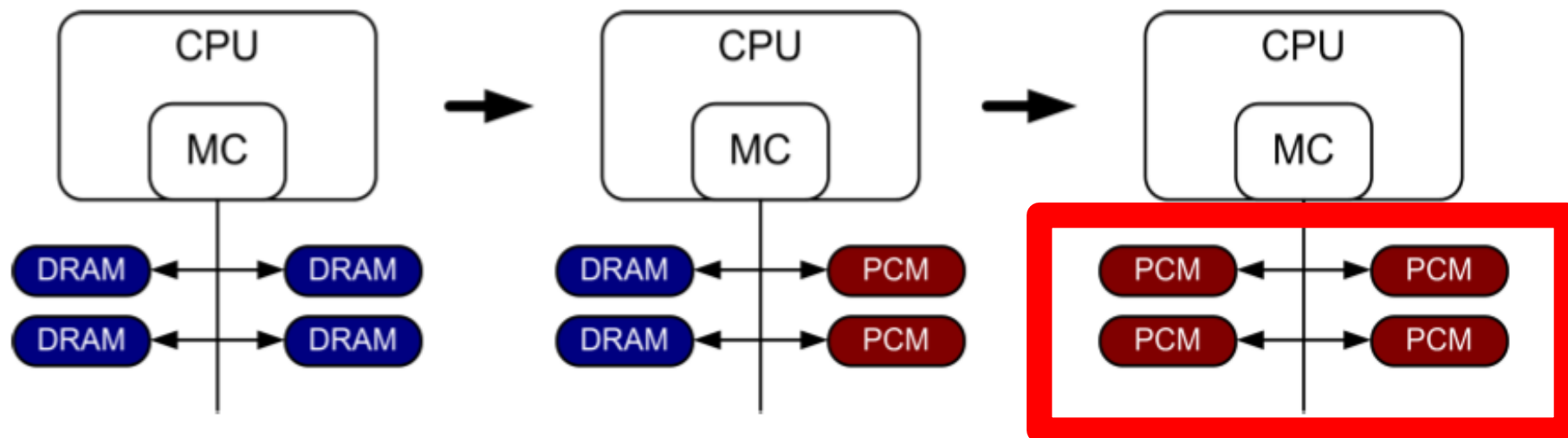
Results: DRAM as PCM Cache (II)

- PCM-DRAM Hybrid performs similarly to similar-size DRAM
- Significant power and energy savings with PCM-DRAM Hybrid
- Average lifetime: 9.7 years (no guarantees)

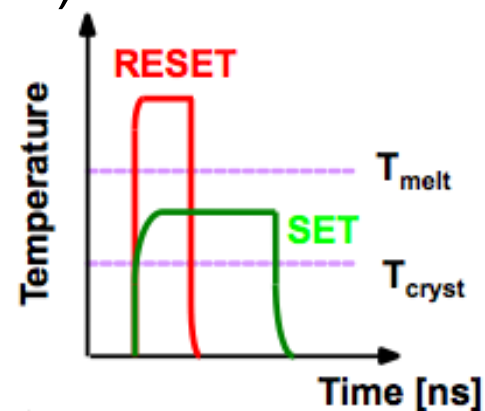


PCM-based Main Memory (II)

- How should PCM-based (main) memory be organized?

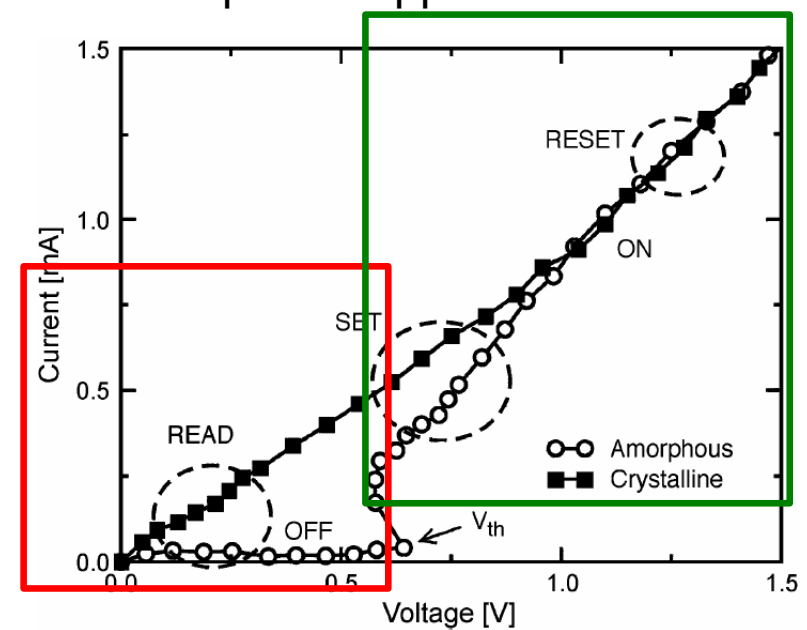


- Pure PCM main memory** [Wang et al., CODES+ISSS'19, Ho et al. NVMSA'21]:
 - How to redesign entire hierarchy (and cores) to overcome PCM shortcomings
 - Worse SET performance
 - Asymmetric write performance
 - Limited endurance (bit-flip numbers)
 - Energy consumption



I-V Characteristics of Set and Reset

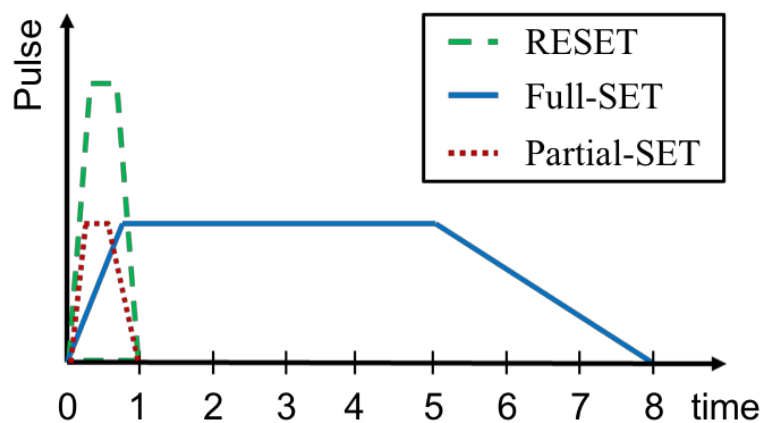
- The set and reset states have large resistance contrast for voltage below V_{th}
 - The reset state (amorphous) is **in the high-resistance state** below V_{th}
 - Very small current ($\sim$ zero)
 - The reset state (amorphous) shows **electronic threshold switching behavior** at V_{th}
 - It is reversible if the voltage pulse is removed very quickly
 - This can be used to partial-SET the PCM cells in some specific applications
 - If the voltage is applied for longer than the crystallization time it leads to **memory switching and the cell reaches the low-resistance state** for an applied voltage larger than V_{th}



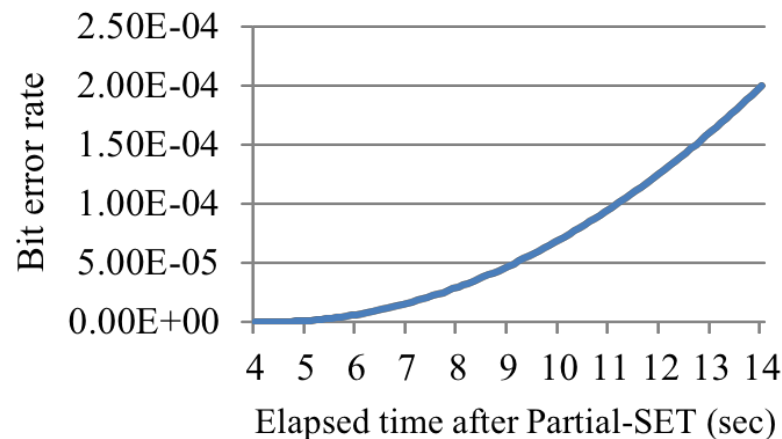
How to Resolve Longer SET performance?

- Compromise between Performance and Data Retention Time - Lossy Programming (Lossy-SET)
- Example operations are Partial-SET [DATE'14] and Quick-and-Dirty [ICCD'17]

Partial-SET vs. Full-SET



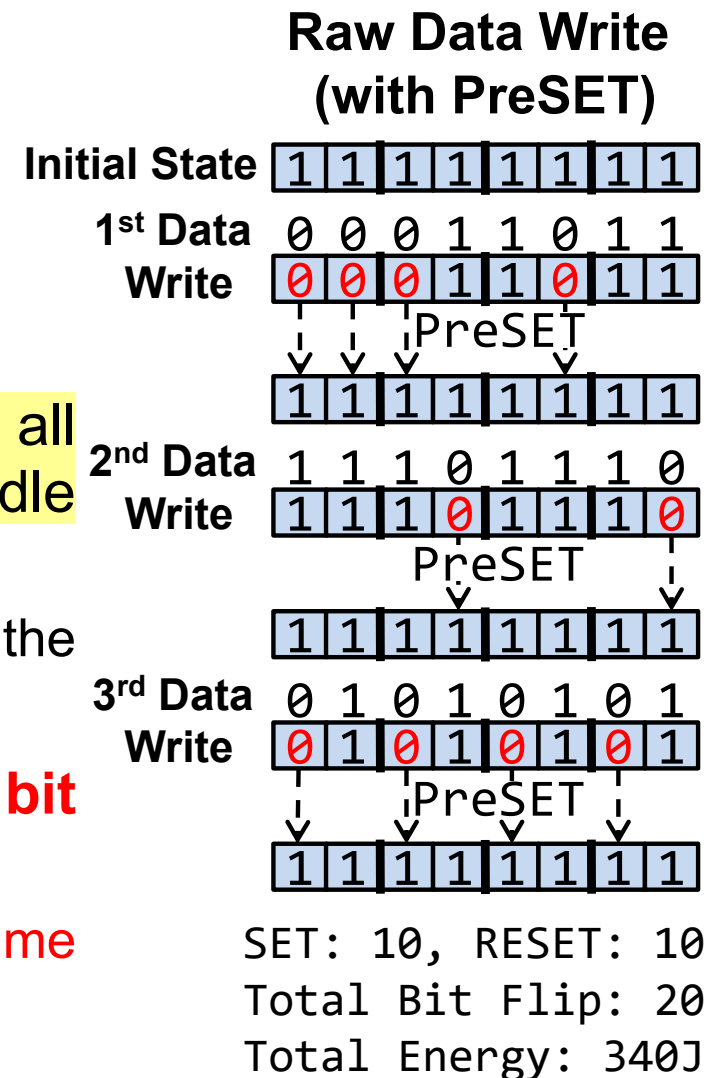
Bit error rate of Partial-SET increases dramatically



Is it possible to train neural networks efficiently on NVM-based systems with Lossy-SET operations but without losing the accuracy?

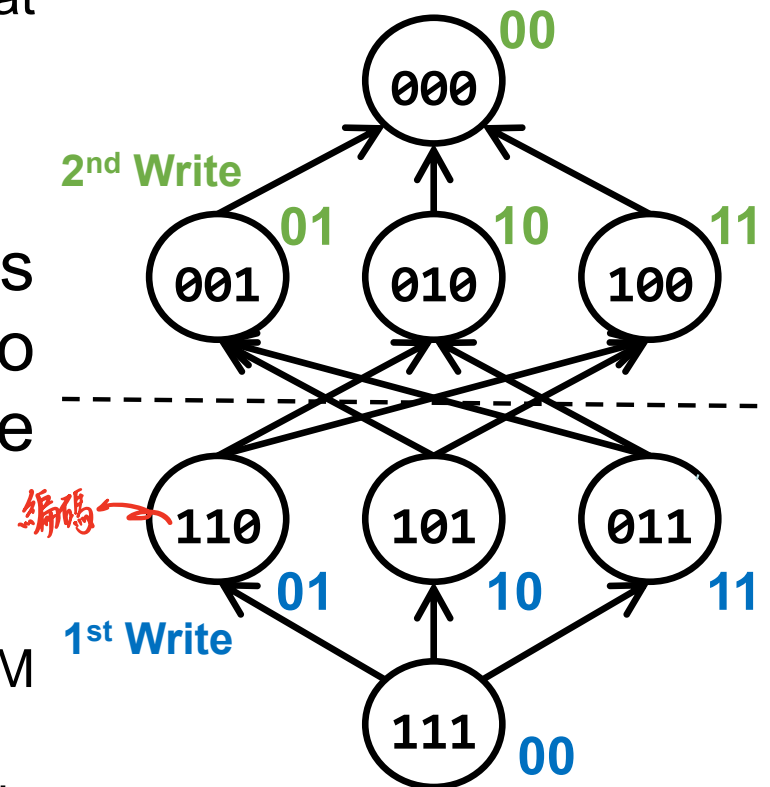
How to Resolve Write Asymmetric Issue of PCM?

- The **write asymmetric issue** is due to the two write operations
 - RESET program PCM cells from '1' to '0' **快**
 - Shorter write latency but **more energy consumption**
 - SET programs PCM cells from '0' to '1' **慢**
 - Longer write latency but **less energy consumption**
- **"PreSET"** proposes to **proactively invoke SET all the data bits into '1's during memory bank idle period**
 - It only executes RESET operation during the memory write period
- PreSET however generates **a large number of bit flips** on PCM cells
 - It thus incurs the **energy consumption** and **lifetime** issues



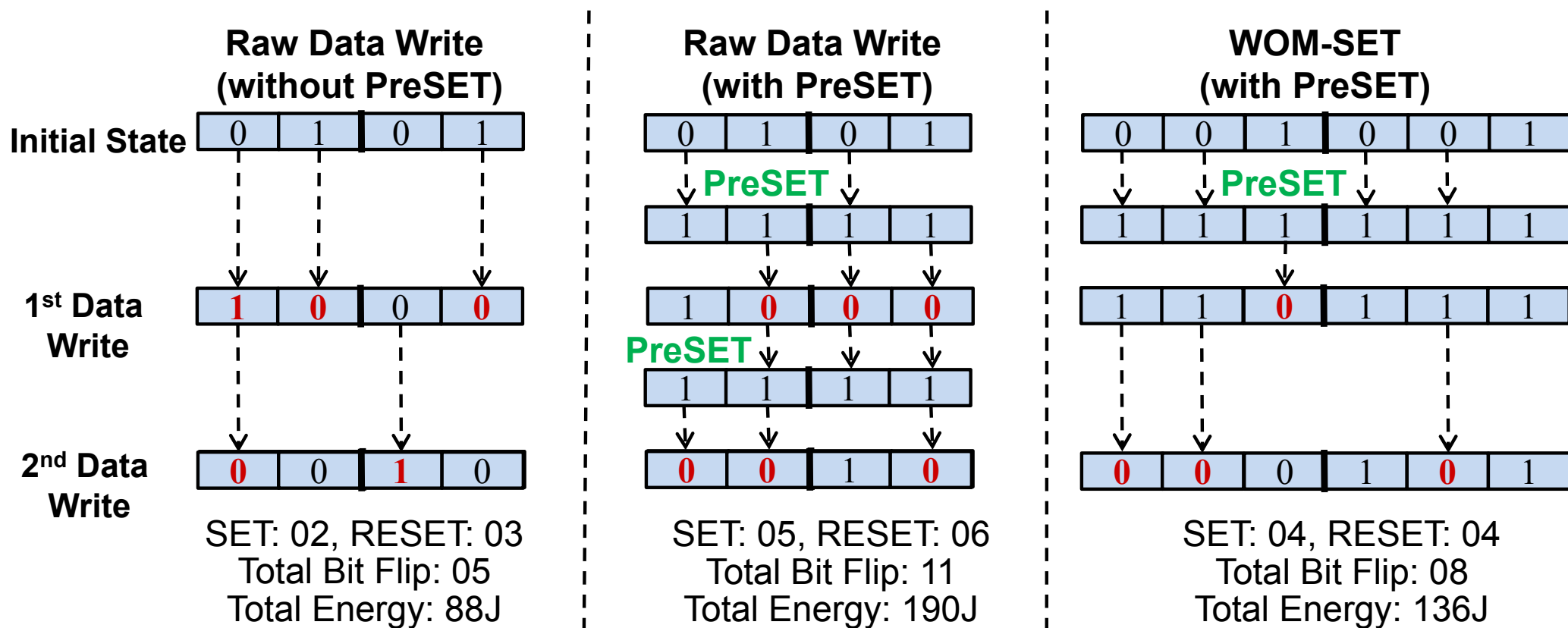
How to Resolve Bit-Flip Unleveling Issue of PCM?

- Write-once memory (WOM) code
 - A “ $\langle v \rangle^t/n$ WOM-code” is a coding scheme that uses n bits to represent one of v values so that the WOM can be written in a total of t times
- WOM-SET combines both advantages of WOM code and PreSET method to improve not only write performance but also write energy efficiency
 - Encode every two-bit data into three-bit data
 - Program the three-bit encoded data on PCM cells *每次只有1 bit會動*
 - Apply PreSET to cells after the 2nd write is finished



$\langle 2 \rangle^2/3$ WOM-code Example

Comparison on Energy and Bit-flips

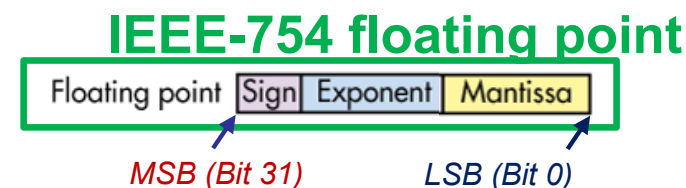


- Baseline approach provides an energy-efficient solution, but it encounters the **uneven bit-flip distribution** issue
- WOM-SET **decreases the number of bit flips** and **limits the energy consumption** without sacrificing the write performance, compared to the basic and PreSET approaches
- WOM-SET could result in the space overhead and then lead to the worse endurance problem on PCM

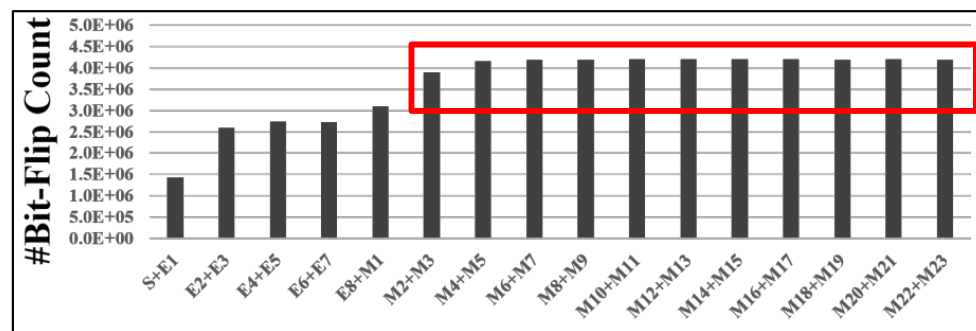
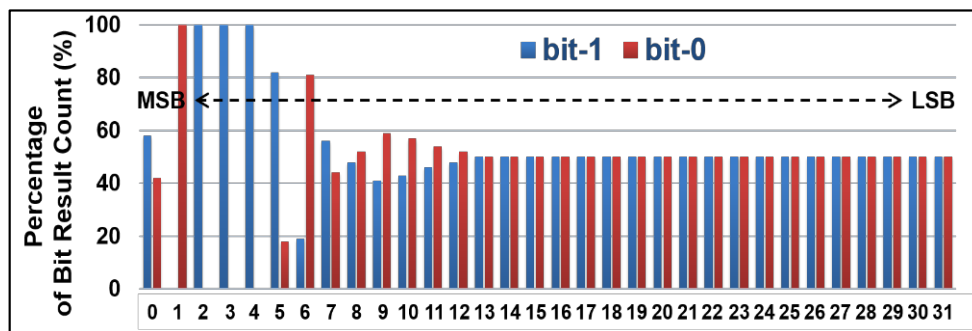
Using WOM-SET on NN Training?

- Statistic of bit result of weights and biases when DenseNet-BC is trained on CIFAR-10 dataset, and it is found that

- The bit-flip on MSBs accumulates **unevenly**
- The bit-flip on LSBs accumulates **very balanced**



- The number of **bit-flip accumulation among encoded LSBs are very large** when applying WOM code for training NNs on NVM-based system
- Energy consumption and endurance issues of NVM-based system with adopting WOM code can be still deteriorated



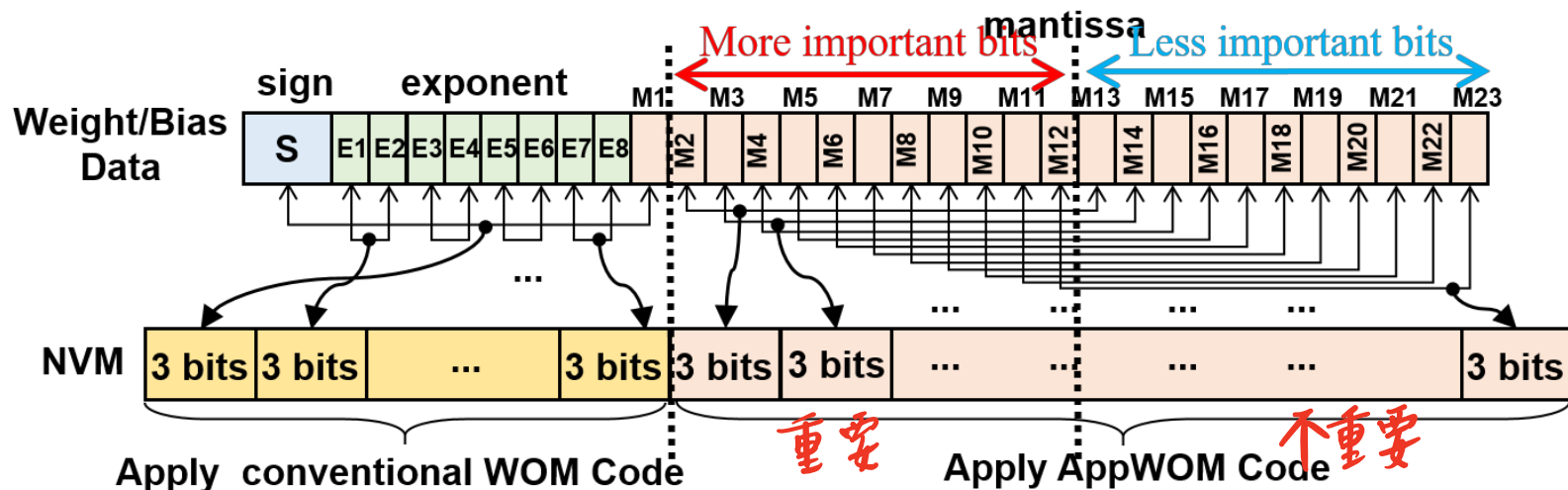
Observation and Motivation

- Observations

- The number of total bit flips is dominated by the mantissa part
- LSBs of weights and biases are more vulnerable to alter
 - The **less important** bits: the latter 11 bits of mantissa
 - The **more important** bits: the former 11 bits of mantissa

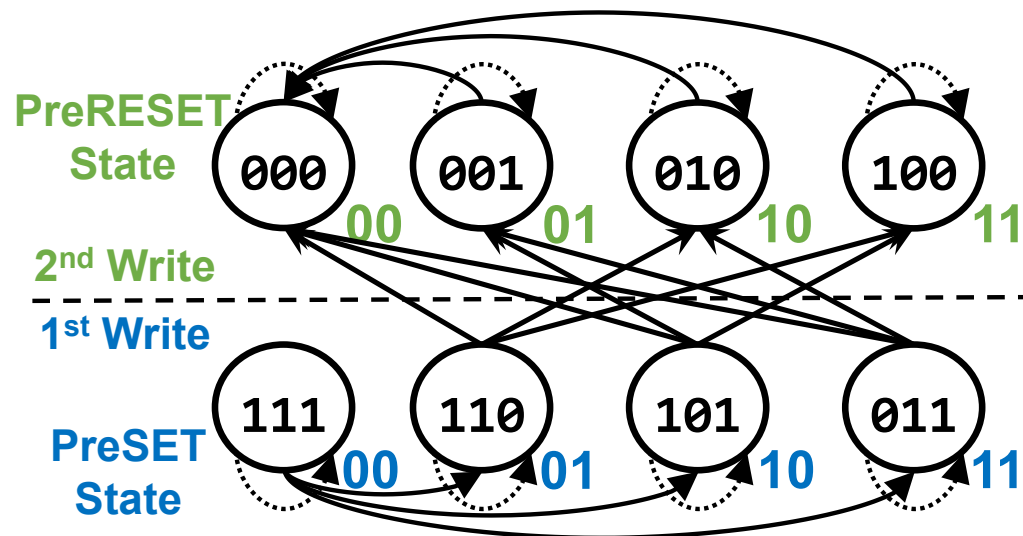
- Bits of weights and biases are partitioned, regrouped, and stored on the NVM

- The first bit of both groups are grouped together, the second bit of both groups are grouped together, and so on



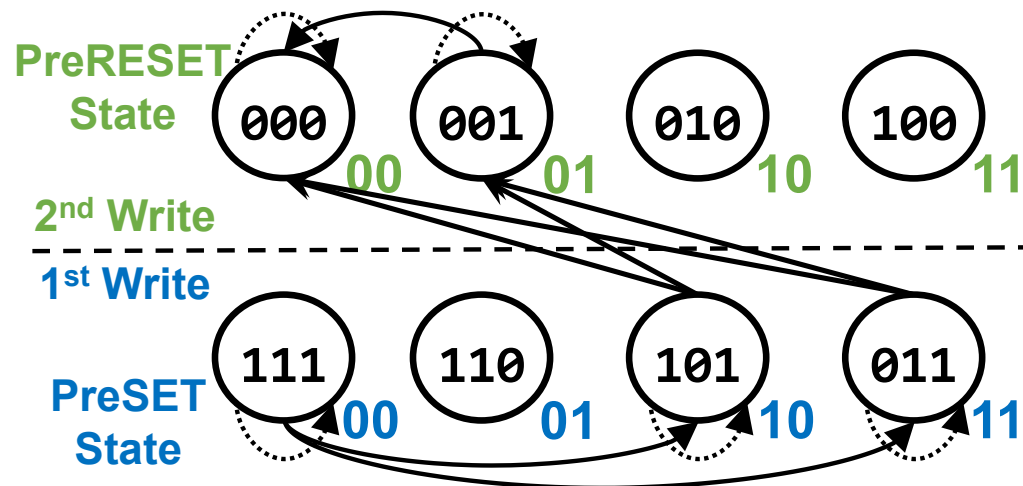
AppWOM Code Method

- AppWOM **updates the encoding state** for the encoded group only when the important bit or both of two bits need to be updated
 - It **keeps the encoding state unchanged** and ignore the update request when it only needs to update the unimportant bit of encoded groups
 - Some path of original WOM code can be discarded according to the above rules
- AppWOM code method can be derived by the following steps:
 - 0. Original state:



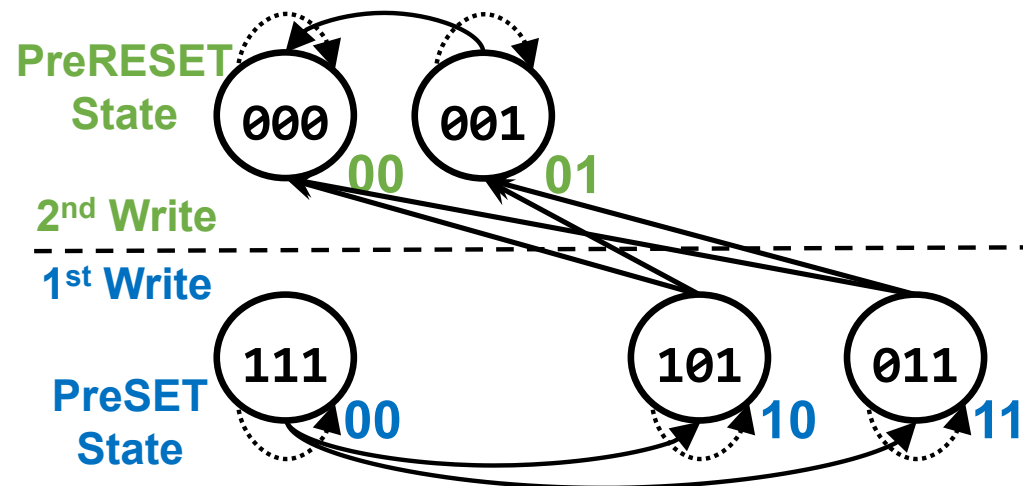
AppWOM Code Method

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 - It **keeps the encoding state unchanged** and ignore the update request when it only needs to update the unimportant bit of encoded groups
 - Some path of original WOM code can be discarded according to the above rules
- AppWOM code method can be derived by the following steps:
 - 1. Remove the paths which represent the change on only the unimportant bit:



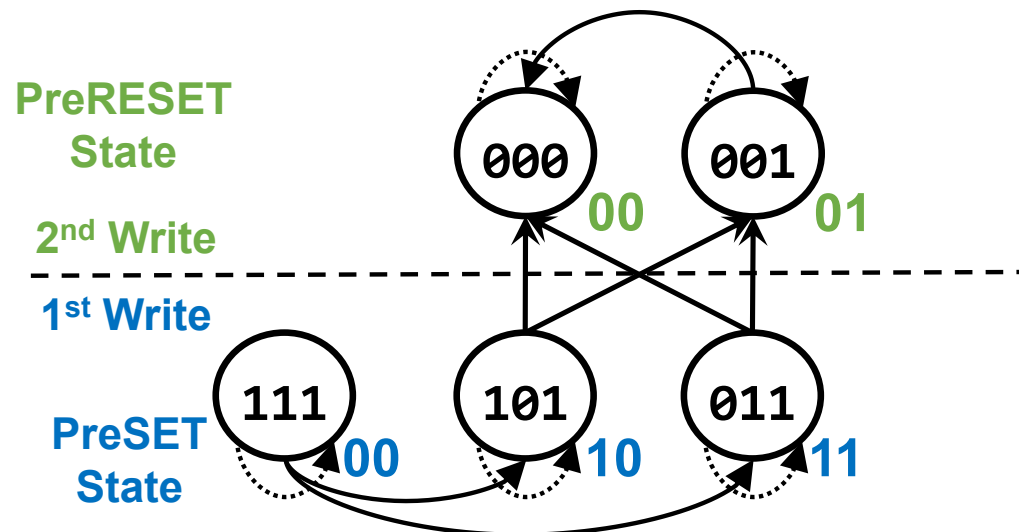
AppWOM Code Method

- AppWOM **updates the encoding state** for the encoded group only when the important bit or both of two bits need to be updated
 - It **keeps the encoding state unchanged** and ignore the update request when it only needs to update the unimportant bit of encoded groups
 - Some path of original WOM code can be discarded according to the above rules
- AppWOM code method can be derived by the following steps:
 - 2. Remove the unused states:



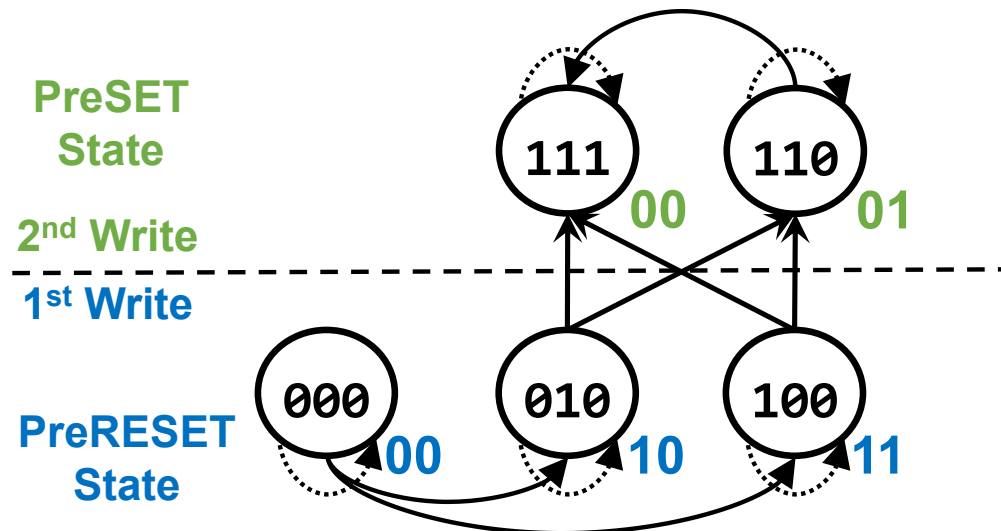
AppWOM Code Method

- AppWOM **updates the encoding state** for the encoded group only when the important bit or both of two bits need to be updated
 - It **keeps the encoding state unchanged** and ignore the update request when it only needs to update the unimportant bit of encoded groups
 - Some path of original WOM code can be discarded according to the above rules
- AppWOM code method can be derived by the following steps:
 - 3. Rearrange states to achieve **AppWOM₁₀**:
 - AppWOM₁₀ is used to serve updates on PCM cells being applied with **PreSET** operations, i.e., all bits are '1'.



AppWOM Code Method

- AppWOM **updates the encoding state** for the encoded group only when the important bit or both of two bits need to be updated
 - It **keeps the encoding state unchanged** and ignore the update request when it only needs to update the unimportant bit of encoded groups
 - Some path of original WOM code can be discarded according to the above rules
- AppWOM code method can be derived by the following steps:
 - 4. Reverse direction to achieve **AppWOM₀₁**:
 - AppWOM₀₁ is used to serve updates on PCM cells being applied with **PreRESET** operations, i.e., all bits are '0'



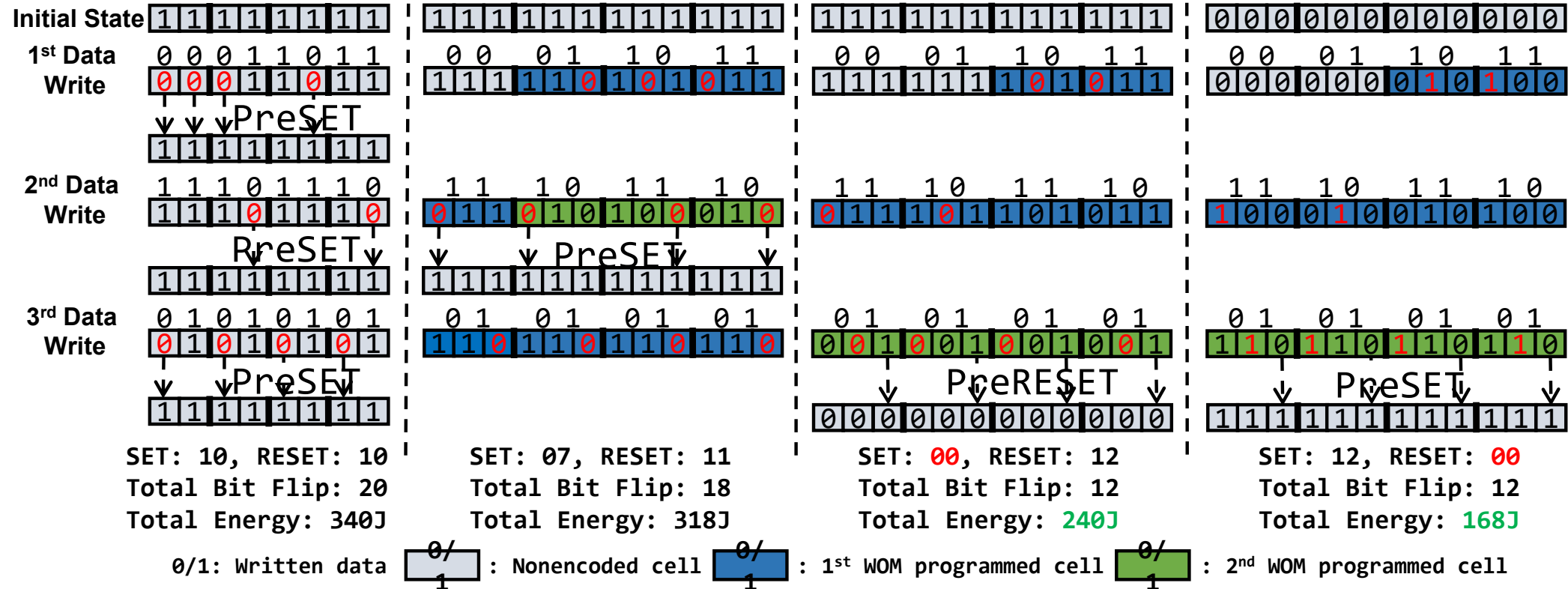
Example of Using AppWOM Code

Raw Data Write (with PreSET)

WOM (with PreSET)

AppWOM₁₀ (with PreSET)

AppWOM₀₁ (with PreRESET)



Agenda

- Overview on the main memory system
- Opportunity: Emerging Memory Technologies
 - Background
 - Phase Change Memory
 - PCM (or Technology X) as DRAM Replacement
 - Hybrid Memory Systems
- Conclusion

Conclusion → PCM 有機會取代 DRAM (在 mem memory)

- PCM has special characteristics and is a good candidate in the designs of different memory hierarchies with challenges:
 - Expensive writes in terms of latency and energy
 - Low rewrite cycles and asymmetric read/write speed